Kuliah Dosen Praktisi

External Internal Lightning Protection for Photo Voltaic Solar Plant

Truly Trust
Commitment

IKATAN AHLI TEKNIK KETENAGALISTRIKAN INDONESIA
IATKI

Unibraw, 30 Oktober 2026

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UNIVERSITAS BRAWIJAYA

0. PENDAHULUAN

Photo Voltaic Plant, PLTS dan Petir

- Indonesia dikaruniai sumber daya alam melimpah, tanah sangat luas dengan sinar matahari tercurah di seluruh negeri, mendukung diwujudkannya kemandirian energi dan ketahanan pangan,
- Potensi dibangunnya PV-Plant, PLTS, sangat besar dan pemerintah telah menetapkan akan dibangunnya 100 GW PLTS.
- Namun Indonesia juga dikenal banyak petir, dimana sambaran petir langsung pada panel surya yang terekspose terbuka menyebabkan kerusakan fatal.
- Induksi medan elektrostatis awan bermuatan akan menyebabkan PID Potential Induced Degradation yang menurunkan umur dan kualitas PLTS.
- Induksi medan elektromagnetis arus sambaran petir menyebabkan kerusakan fatal.

Premature Aging,
Penuaan dini pada komponen PLTS

- Indikasi telah terjadinya Premature Aging: sering eror, overheat, efisiensi turun,
- Premature aging terutama disebabkan oleh PID-Potential Induced Degradation
- Terjadinya PID terutama karena sambaran petir daan induksi medan elektrostatis awan bermuatan.
- Tegangan Tinggi bikin arus bocor dari sel ke frame, daya panel bisa turun 30-50% dalam 2 tahun. Umur panel surya: 25-30 tahun
- Medan Ele.ktrostatis: musuh senyap menyebabkan Panel Surya melemah perlahan karena PID.
- Medan Elektromagnetik musuh frontal dapat menyebabkan Panel Surya pecah, Inverter meledak
- Keduanya wajib diproteksi. Biaya external lightning protection, Grounding dan SPD Surge Protection Cuma 1-2% CAPEC,

I. LIGHTNING

Lightning







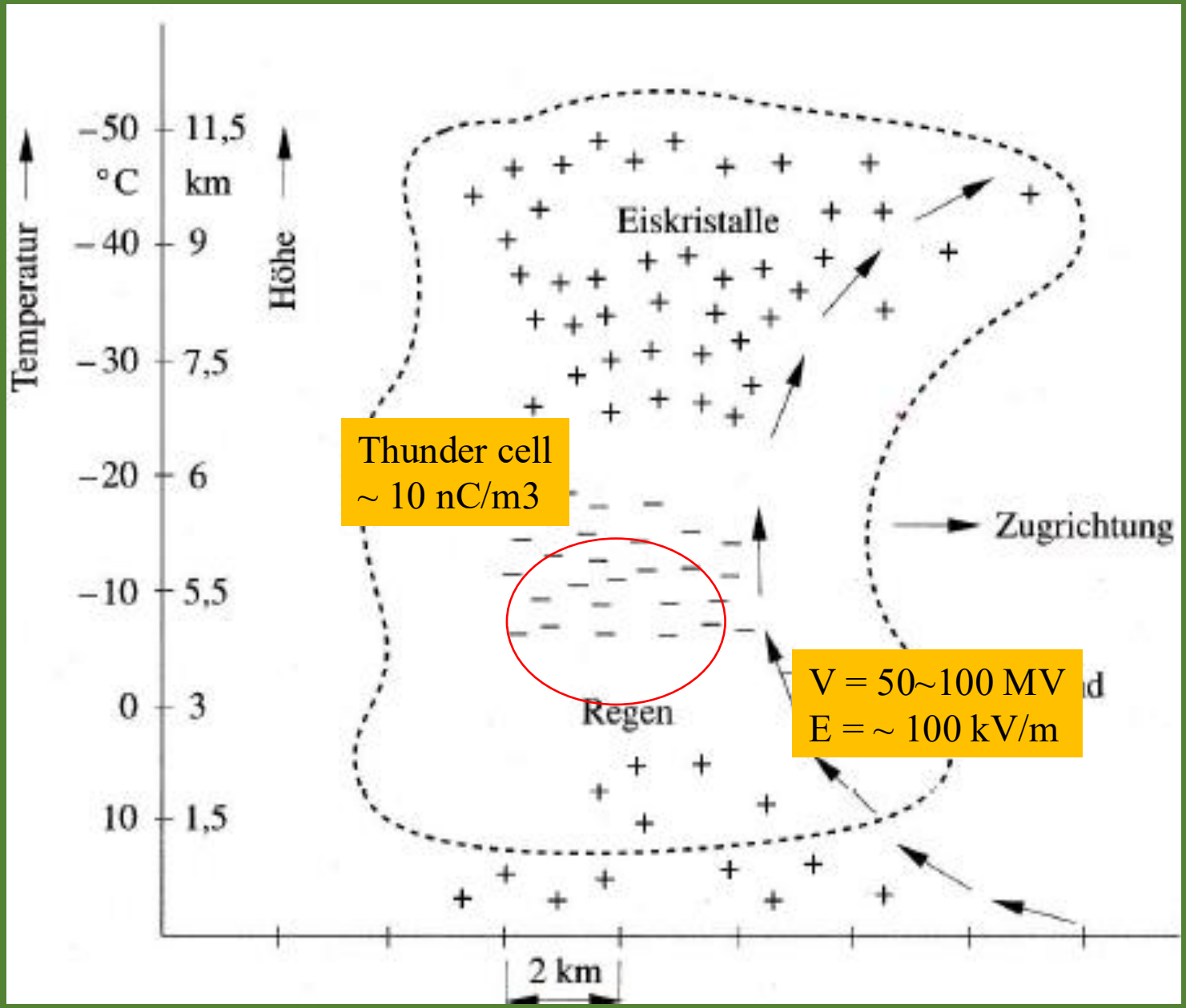
Lightning Strikes Phenomenon



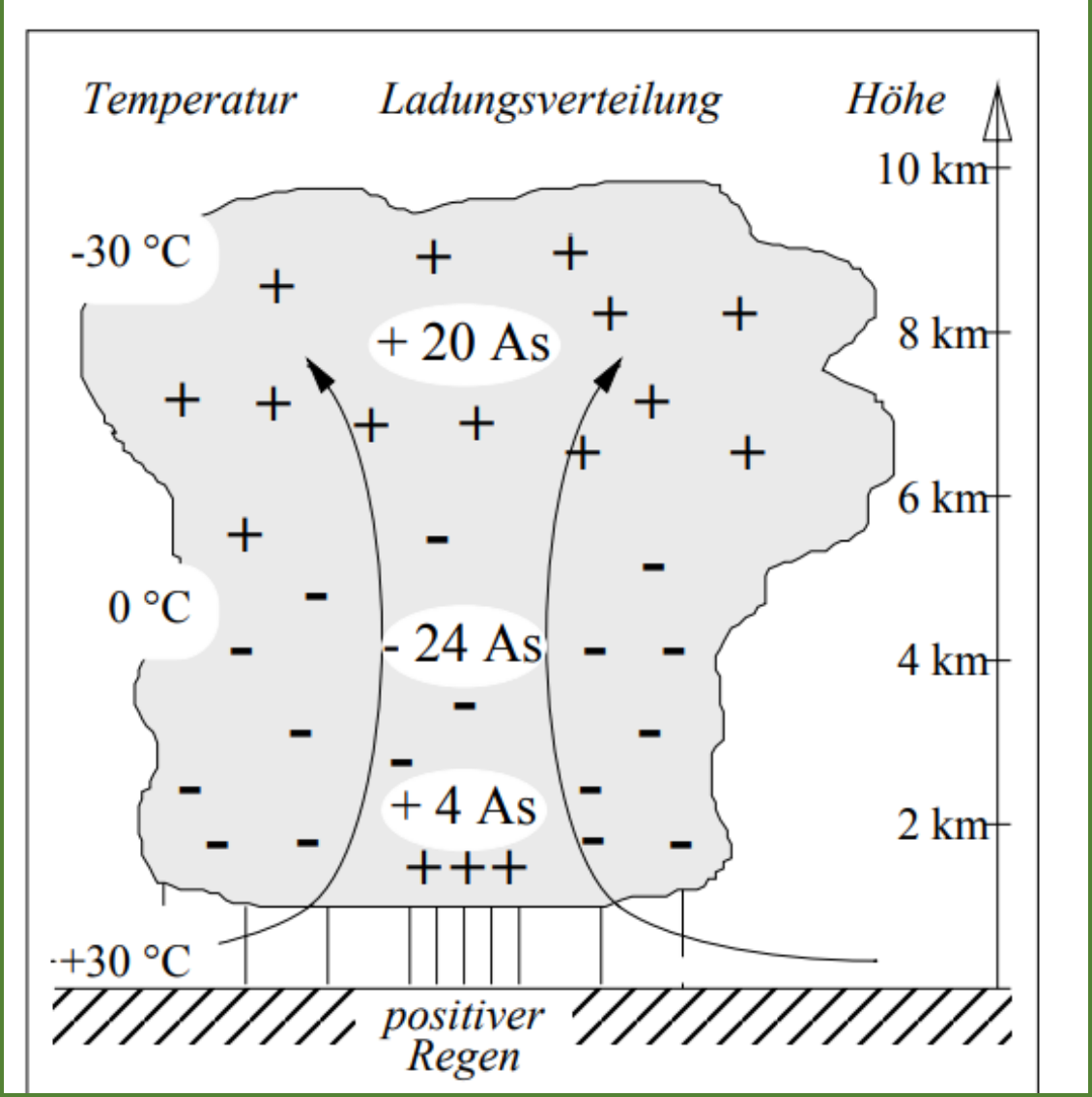
Source: http://www.lightningtech.com/f_sets/facilities.html


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Thunder cells in thunder cloud (Cumulonimbus)



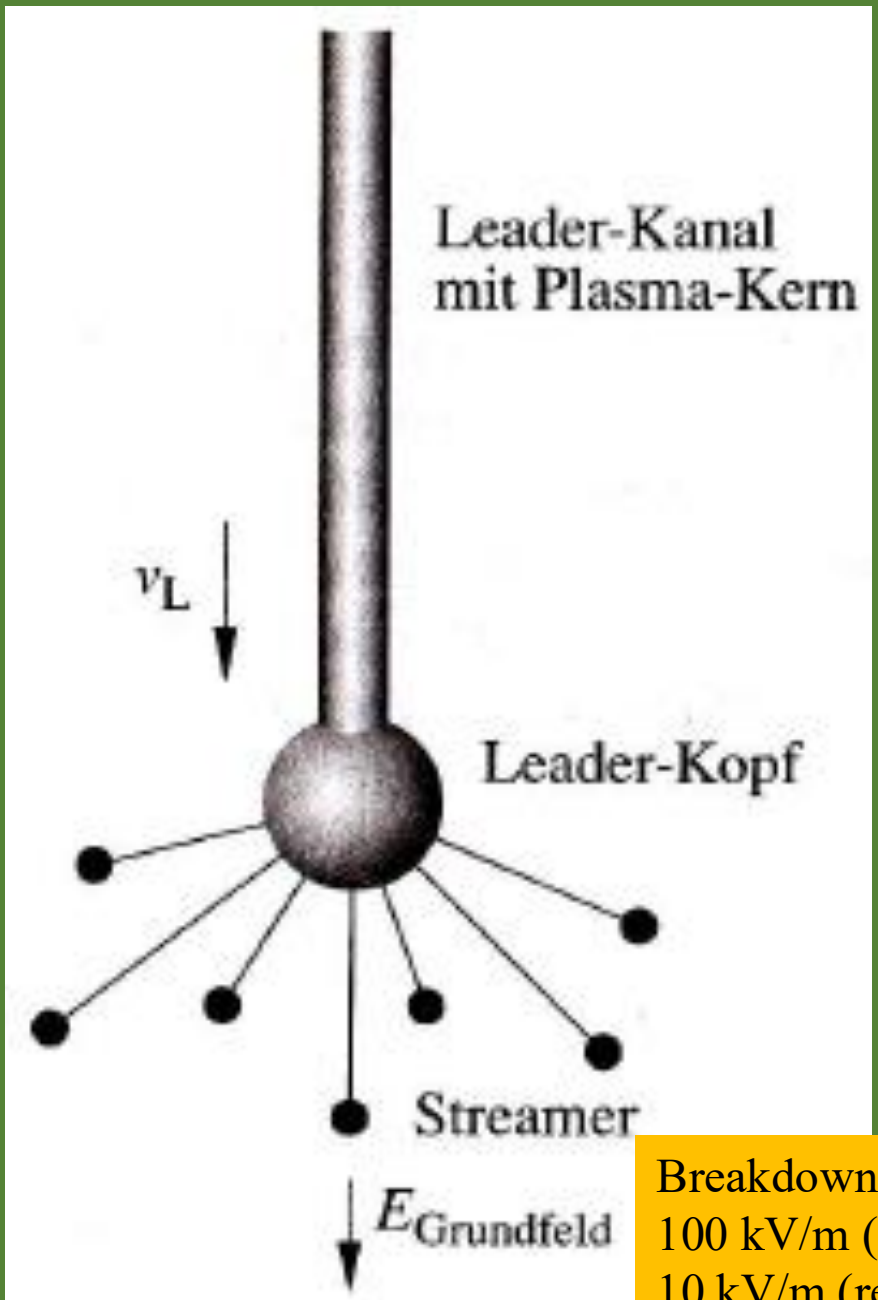
Distribusi muatan pada suatu awan badai

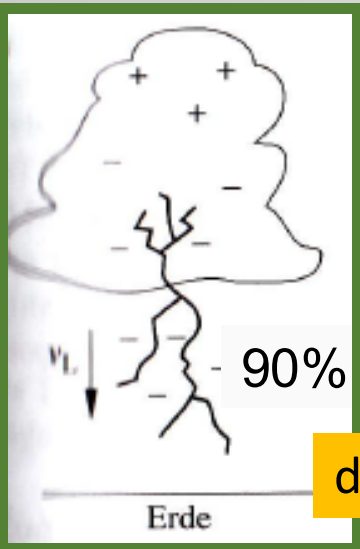


1.1. LIGHTNING STRIKE

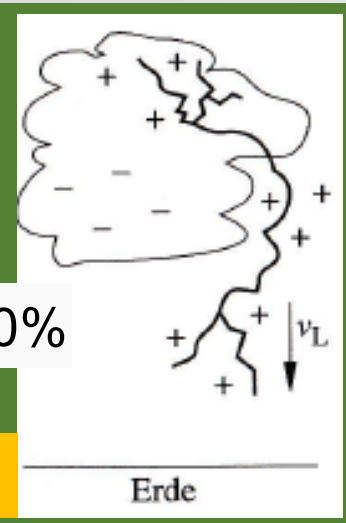


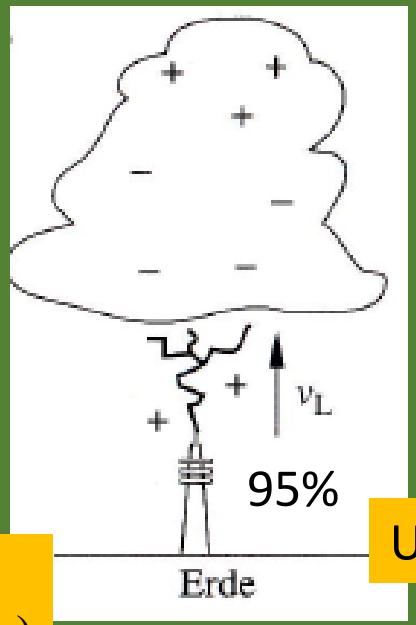
Fisikalis sambaran petir



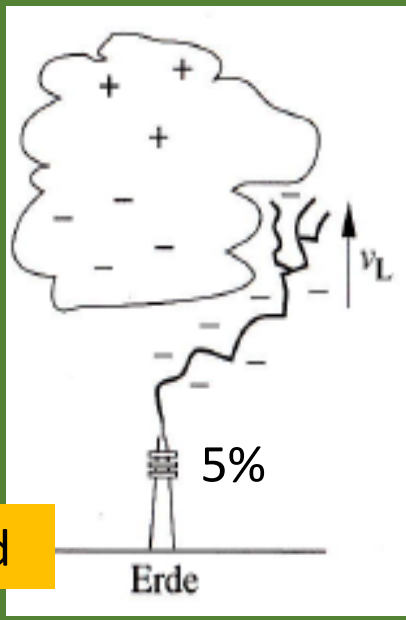


downward



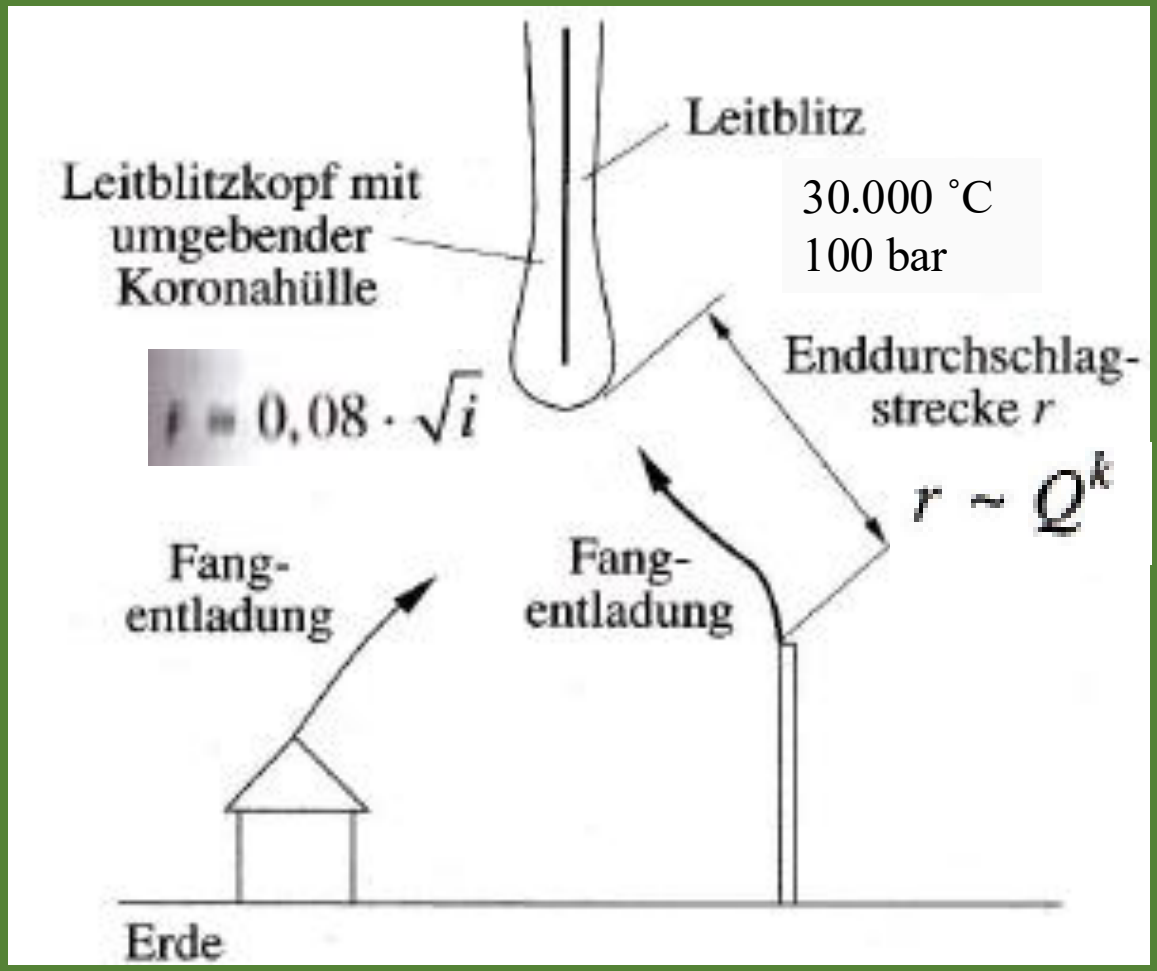


Upward



Breakdown
100 kV/m (labor)
10 kV/m (real)

Mekanisme sambaran petir



$S = 2I + 30\left(1 - e^{-I/6.8}\right)$	Darveniza (1975)
$S = 10\, I^{0.65}$	Love (1987; 1993)
$S = 9.4\, I^{2/3}$	Whitehead (1974)
$S = 8\, I^{0.65}$	IEEE (1985)
$S = 3.3\, I^{0.78}$	Suzuki (1981)

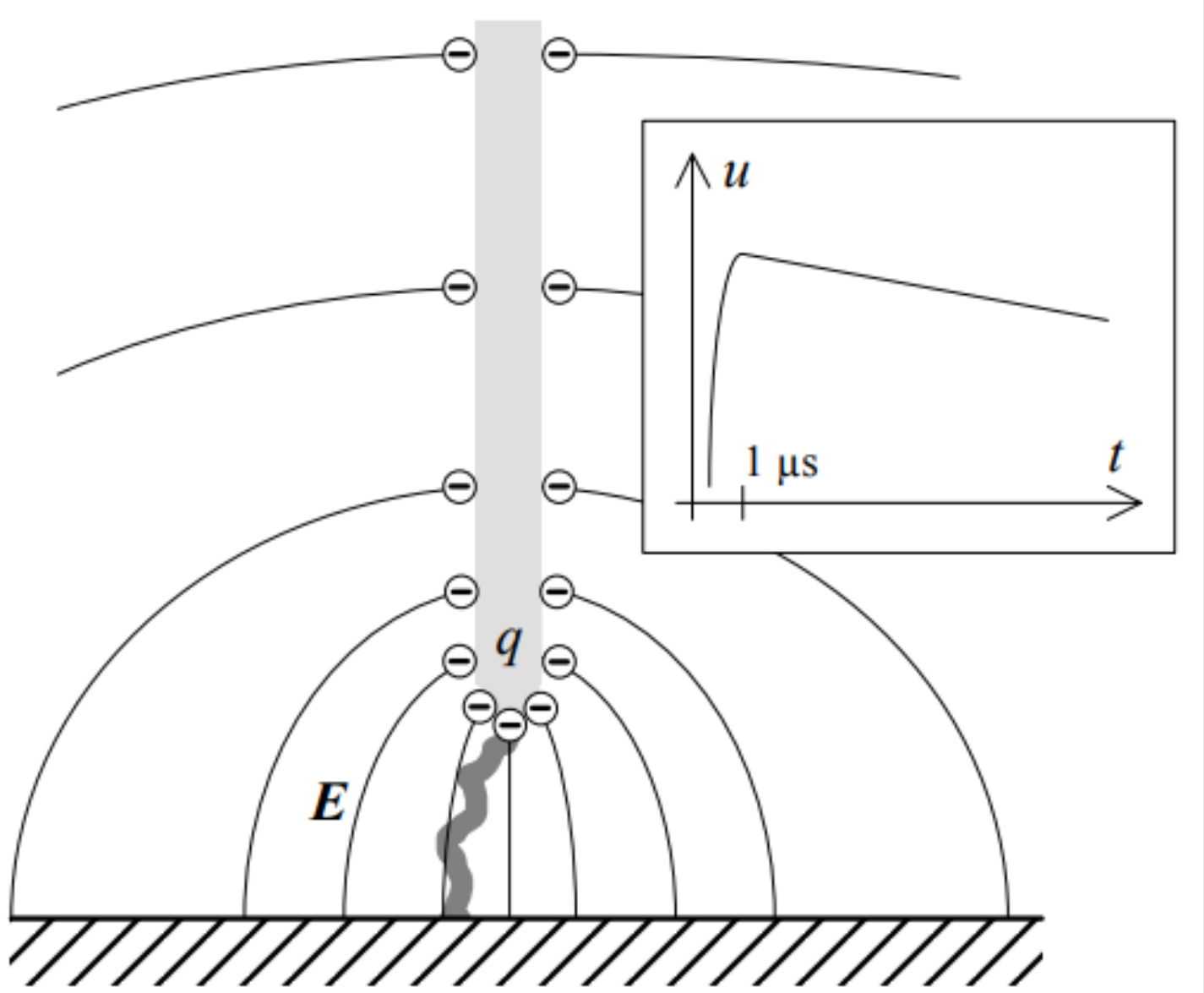
$$r = 10 \cdot i_{\max}^{0,65}$$

$$i_{\max} = 10{,}6 \cdot Q^{0,7}$$



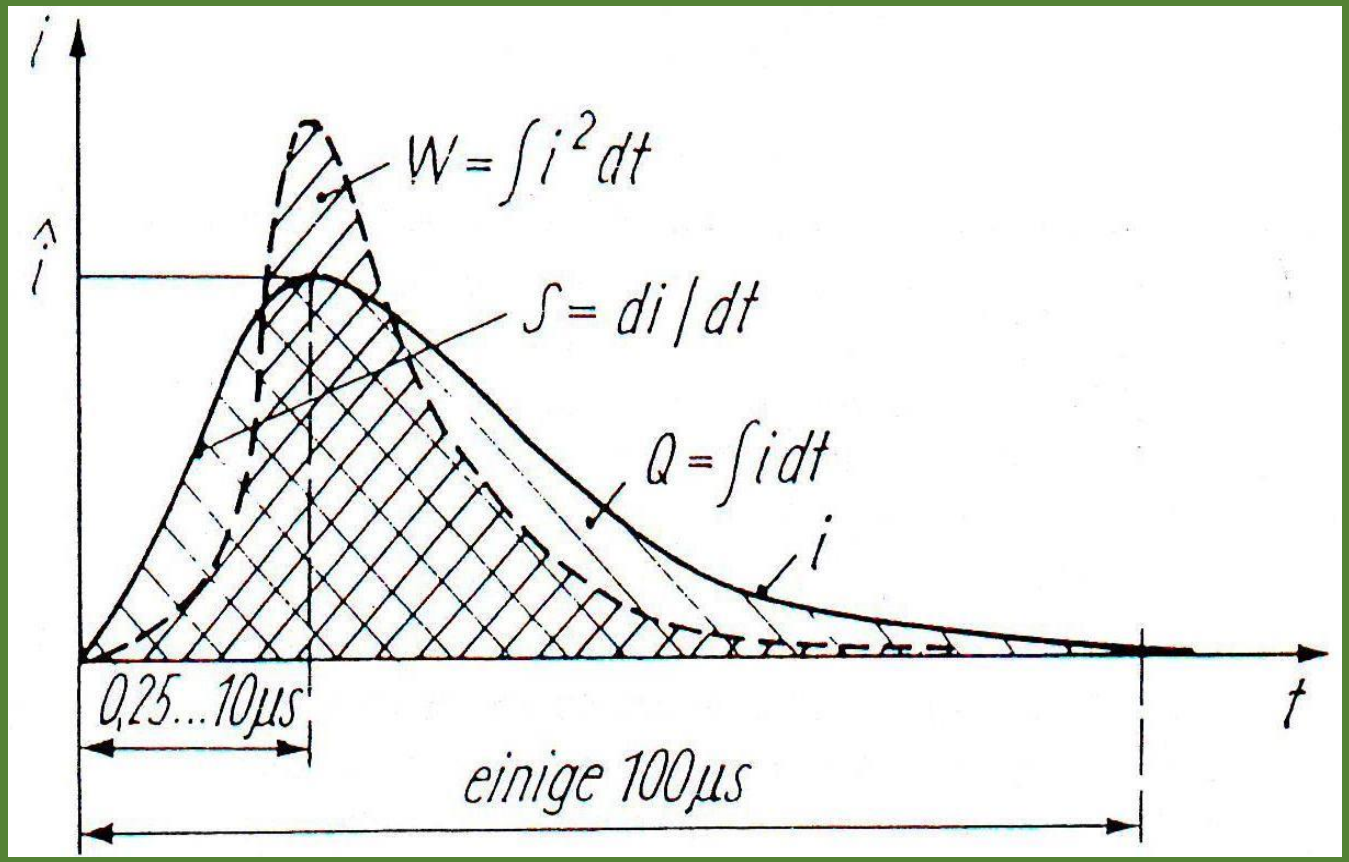


Gambaran sederhana sambaran petir



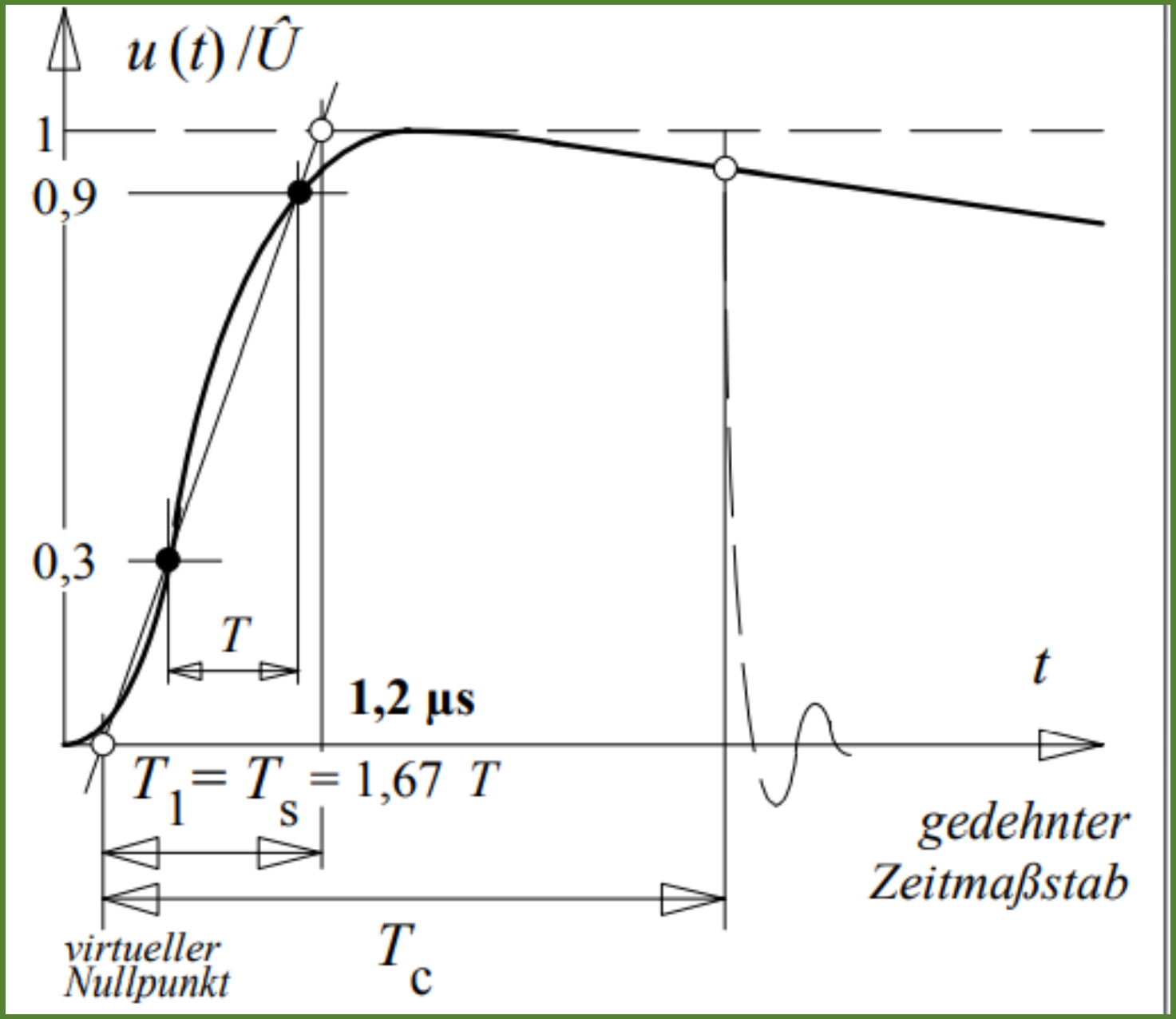
1.2. LIGHTNING IMPULSE PARAMETER

Parameter Arus Petir.



1) Nilai puncak arus I, berakibat tegangan lebih pada tempat sambaran
2) Kecuraman arus S = di/dt, berakibat tegangan induksi elektromagnetis
3) Muatan arus Q = $\int$ i dt, berakibat pindahan energi yang bisa melebur ujung tempat sambaran
4) Energi spesifik arus W/R = $\int$ i^2 dt, berakibat pemanasan dan gaya elektrodinamis pada penghantar

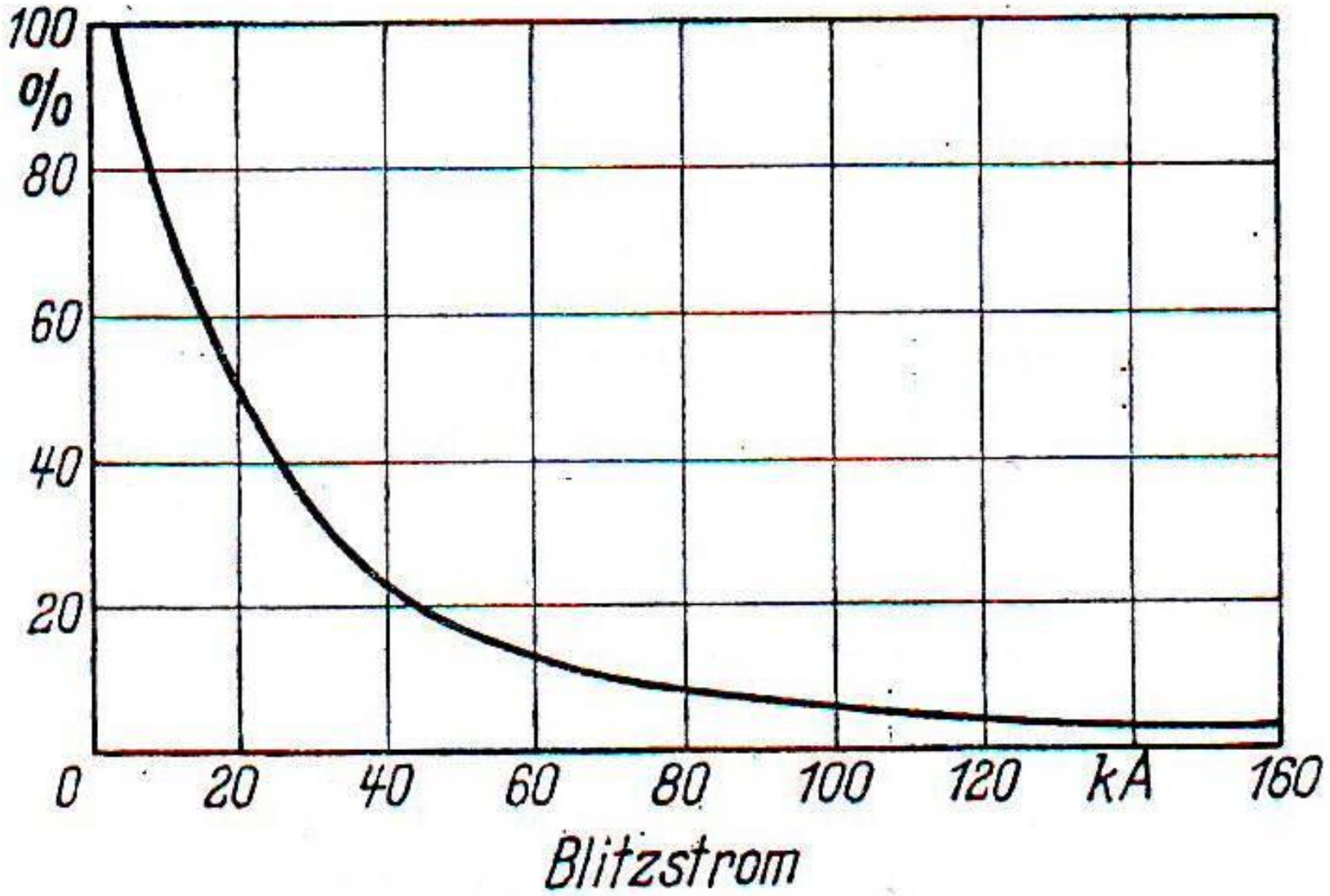
Tegangan Impuls Petir



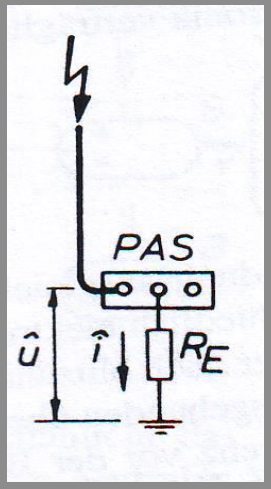
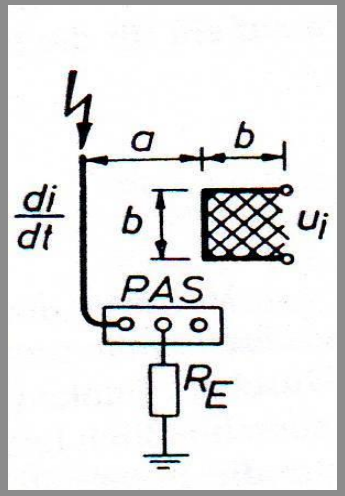
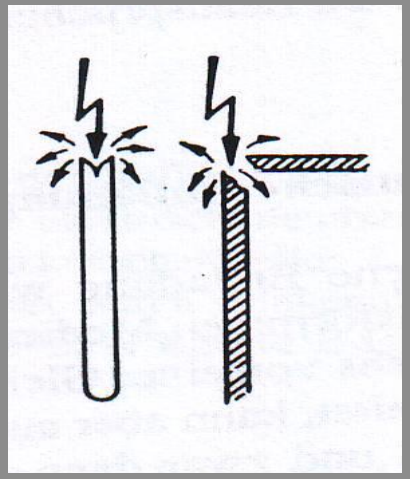
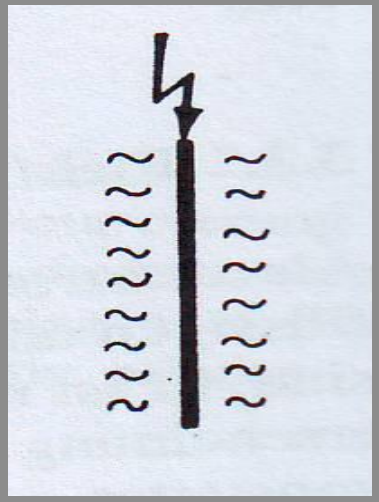
Besaran Parameter Arus Petir

Parameter Petir	Simbol (unit)	Kelas Proteksi		
		I	II	III dan IV
Nilai arus puncak	**I** (kA)	200	150	100
Muatan total	**Q**total (C)	300	225	150
Muatan impuls	**Q**impuls (C)	100	75	50
Energi spesifik	**W/R** (kJ/Ω)	10.000	5.600	2.500
Kecuraman rata-rata	di/dt (kA/μs)	60-180	45-135	30-90

Statistik Besaran Arus Petir

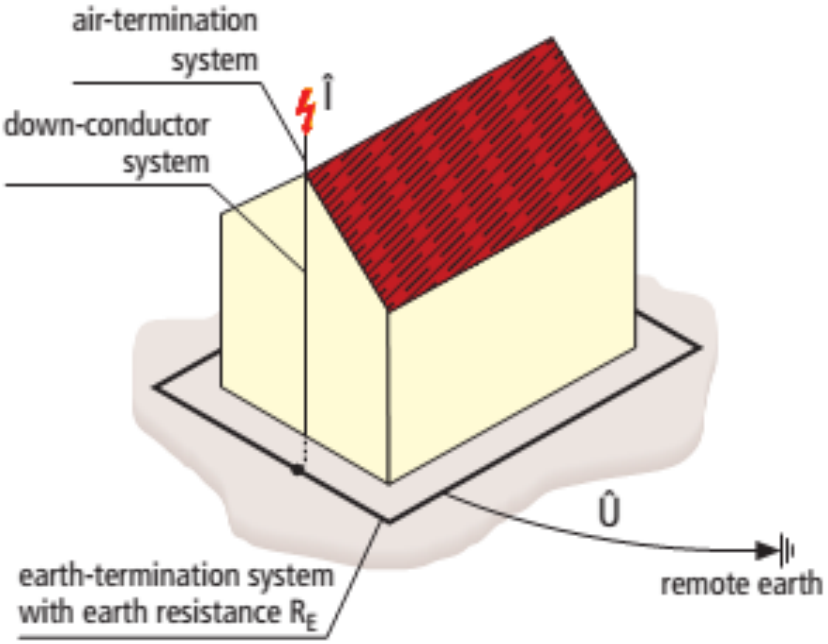


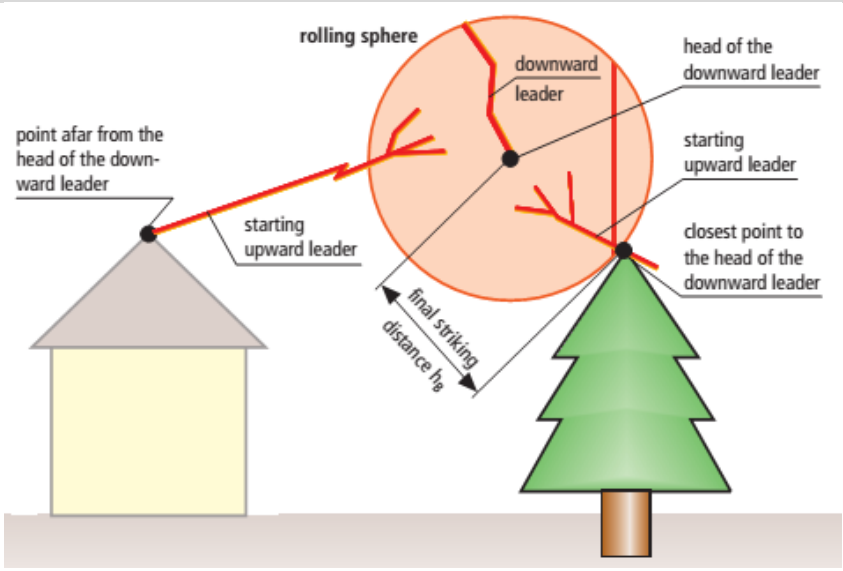
Dampak dari Parameter Arus Petir

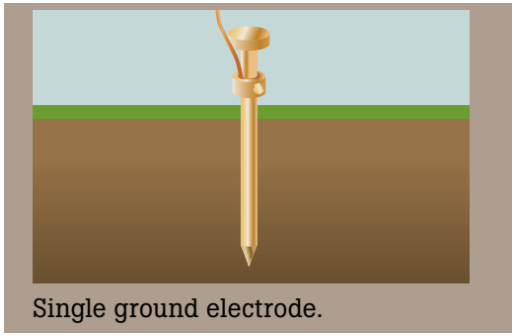
PARAMETER	ARUS PUNCAK $\hat{I}$	KECURAMAN $S = dI/dt$	MUATAN $Q = \int I\,dt$	ENERGI $W/R = \int I^2\,dt$
Besar	2 ... 200 kA	2 ... 200 kA/s	150 .... 3000 As	2,5 ... 10 MJ/Ω
Yang diakibatkan di titik sambar	Kenaikan Tegangan	Tegangan Induksi $U_i = K_u * S$	Pelelehan	Pemanasan Konduktor
				
Jumlah:	î = 200 kA R_E = 5Ω û = 1000 kV	S = 200 kA/µs a = 10 m b = 0,1 m/10 m u_i = 40 V/216 kV	Muatan 300 As dapat melelehkan pelat alumunium tebal 5 mm	Energi petir 10 MJ/Ω melelehkan kawat baja 10 mm^2; kawat tembaga 25 mm^2

II. Air Termination System

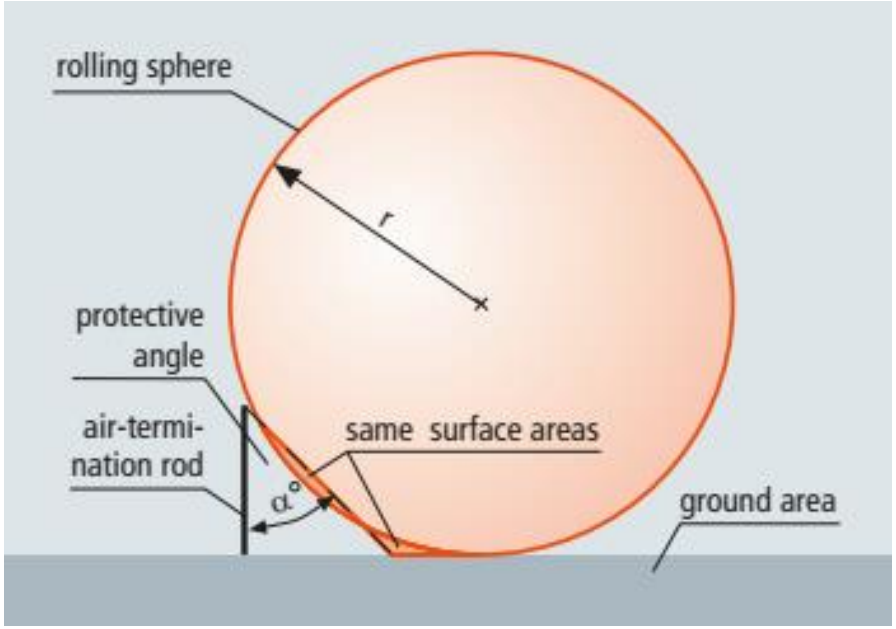
Lightning Strike:
External Lightning Protection






Single ground electrode.

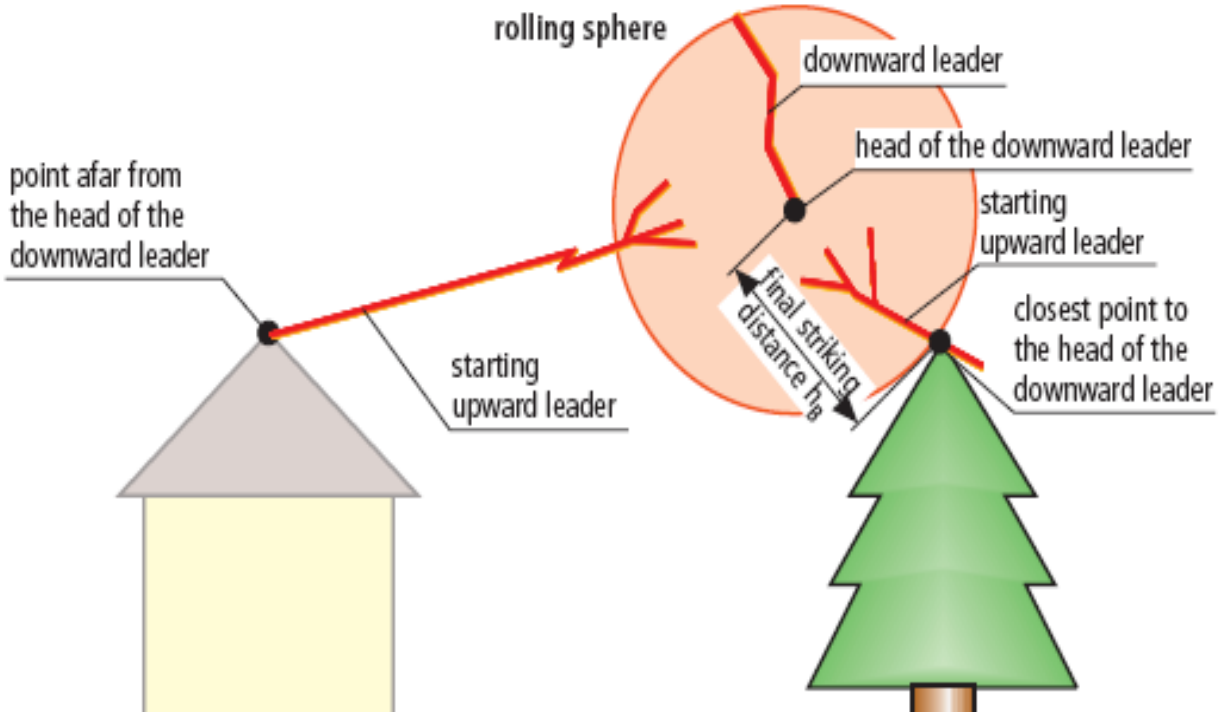




Air Terminal

Roling Sphere Method (Geometric Electrical Model)

Ujung sambaran turun (*downward leader*) mendekati bumi secara bertahap hingga beberapa ratus meter di atas bumi, hingga mencapai jarak sambar terakhir (*final striking distance*). Ujung sambaran turun sambut oleh sambaran naik hingga terjadi peluahan muatan petir ke bumi.



$R = 10 \cdot I^{0,65}$
R (meter) dan I (kA)
Golde

$R = 7,1 \cdot I^{0,75}$
Pathfinder

$R = 6,7 \cdot I^{0,8}$
Amstrong-Whitehead

Natural Air Terminal

Bagian metal dari struktur bisa digunakan sebagai komponen natural dari sistem terminasi-udara. Jika struktur memiliki konstruksi baja dan fasad terbuah dari material konduktif, maka struktur tersebut bisa digunakan selama memenuhi syarat ketebalan minimum.

Ketebalan Minimum Pelat Metal

Bahan	Ketebalan	
	Boleh Meleleh	Tidak Boleh Meleleh
Besi	0,5 mm	4 mm
Tembaga	0,5 mm	5 mm
Aluminium/ Stainless-Steel	0,7 mm	7 mm

Lightning Protection Level

It starts with a first risk analysis in accordance with IEC 62305-2 for the determination of the necessity of protective measures.

A **protection level** I to IV has to be determined in order to define a set of lightning current paramenters values.

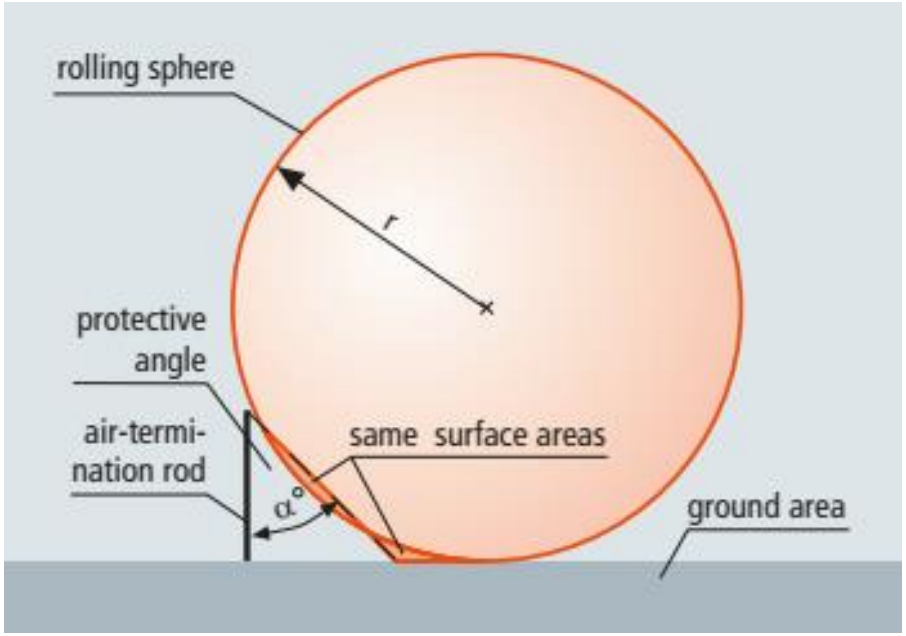
This protection level is used to classify protection measures to define the **type of lightning protection systems (LPS)** needed.

Table of minimum thicknesses of metal sheets for air-termination systems (see IEC 62305-3)

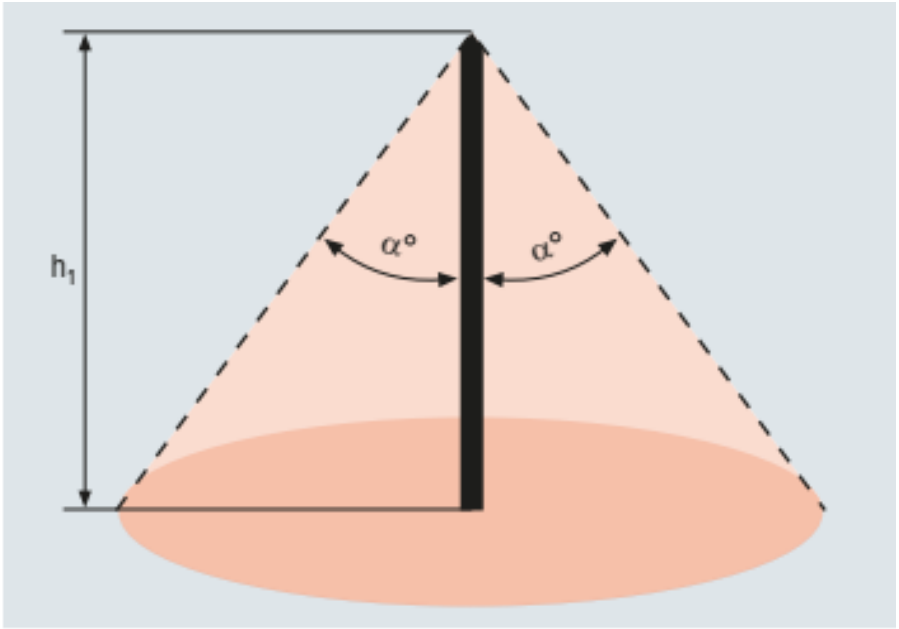
Class of LPS	Material	Thickness t (mm)	Thickness t' (mm)
I to IV	Lead	-	2.0
	Stainless steel or galvanized steel	4	0.5
	Titanium	4	0.5
	Copper	5	0.5
	Aluminium	7	0.65
	Zinc	-	0.7

1.3. Air Terminal Design: *Protective Angle Methode*

Protective Angle Method



Protective angle and comparable radius of the rolling sphere

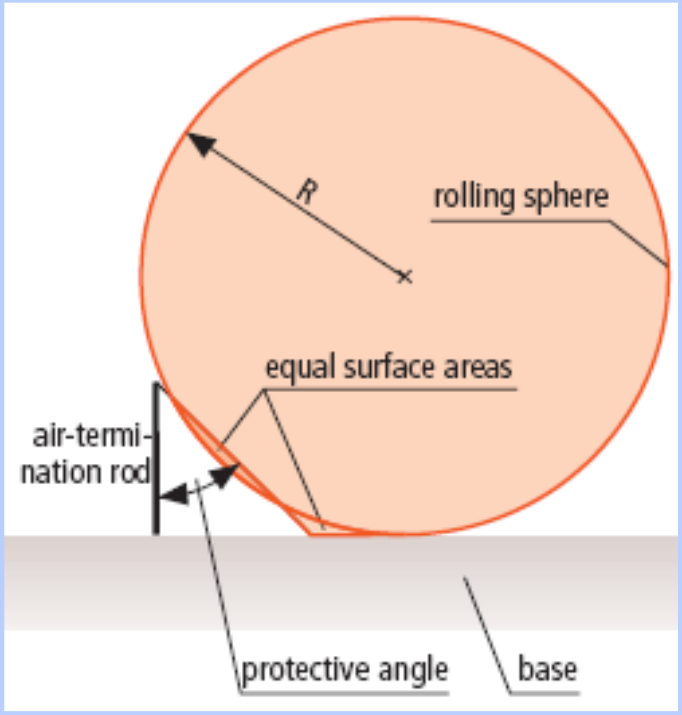


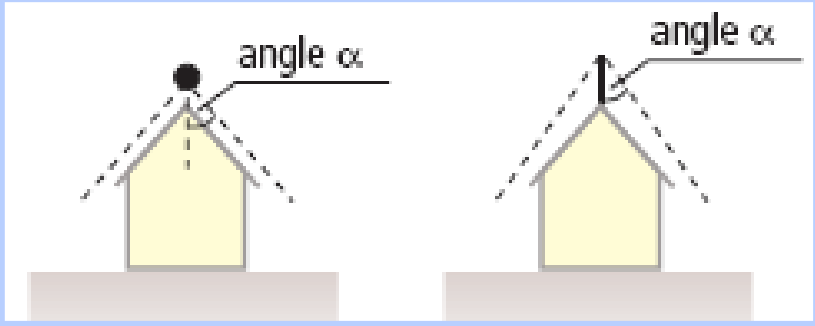
Cone shaped protected volume

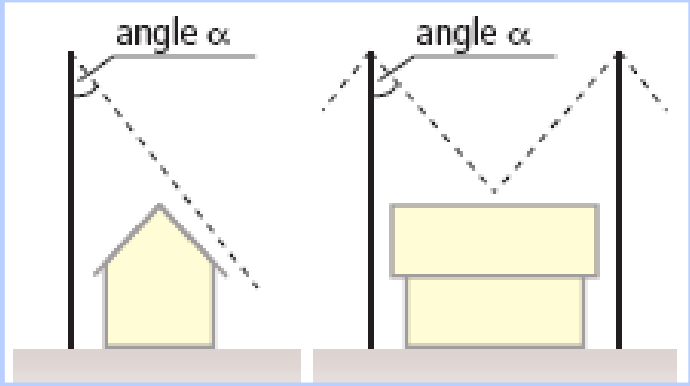
Sistem Terminasi-udara: *Protective Angle*

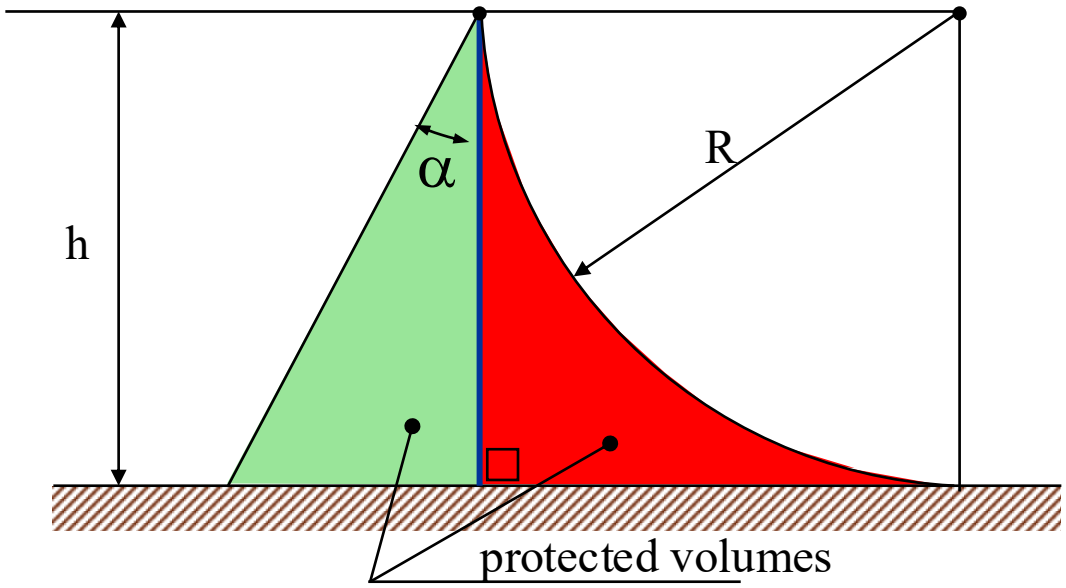
***Protective angle* method** (metode sudut proteksi) diturunkan dari *electrical-geometrical lightning model*. Sudut proteksi ditentukan oleh radius dari bola gulir yang diirisi oleh garis miring sedemikian rupa sehingga diperoleh daerah yang sama besarnya.

Metode ini digunakan untuk struktur simetris atau struktur di atas atap. Sudut proteksi tergantung pada tipe proteksi dari sistem proteksi petir dan tinggi sistem terminasi-udara di atas permukaan rujukan. Zona proteksi bisa berbentuk cone atau tenda.



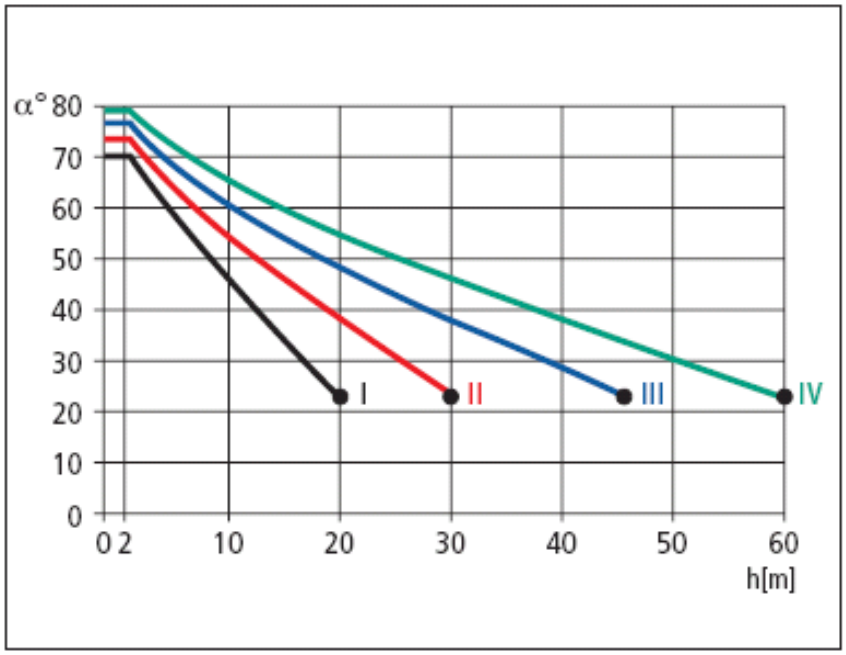




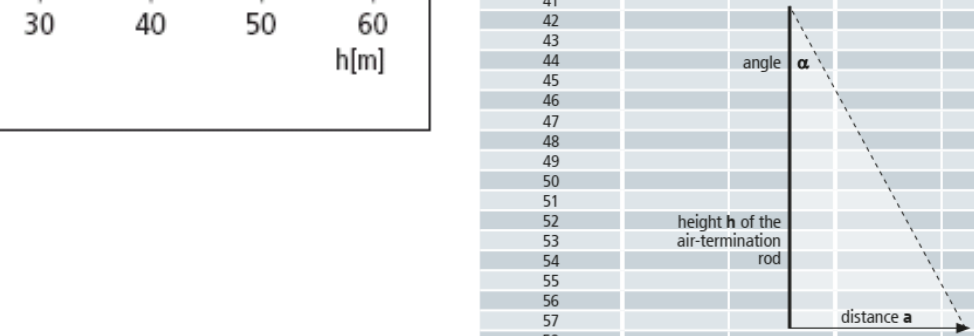


Level	R(m)	α(h = 20)	α(h = 30)	α(h = 45)	α(h = 60)	d(m)
I	20	25	*	*	*	5
II	30	35	25	*	*	10
III	45	45	35	25	*	15
IV	60	55	45	35	25	20

Protective angle α as a function of height h depend on the class of LPS

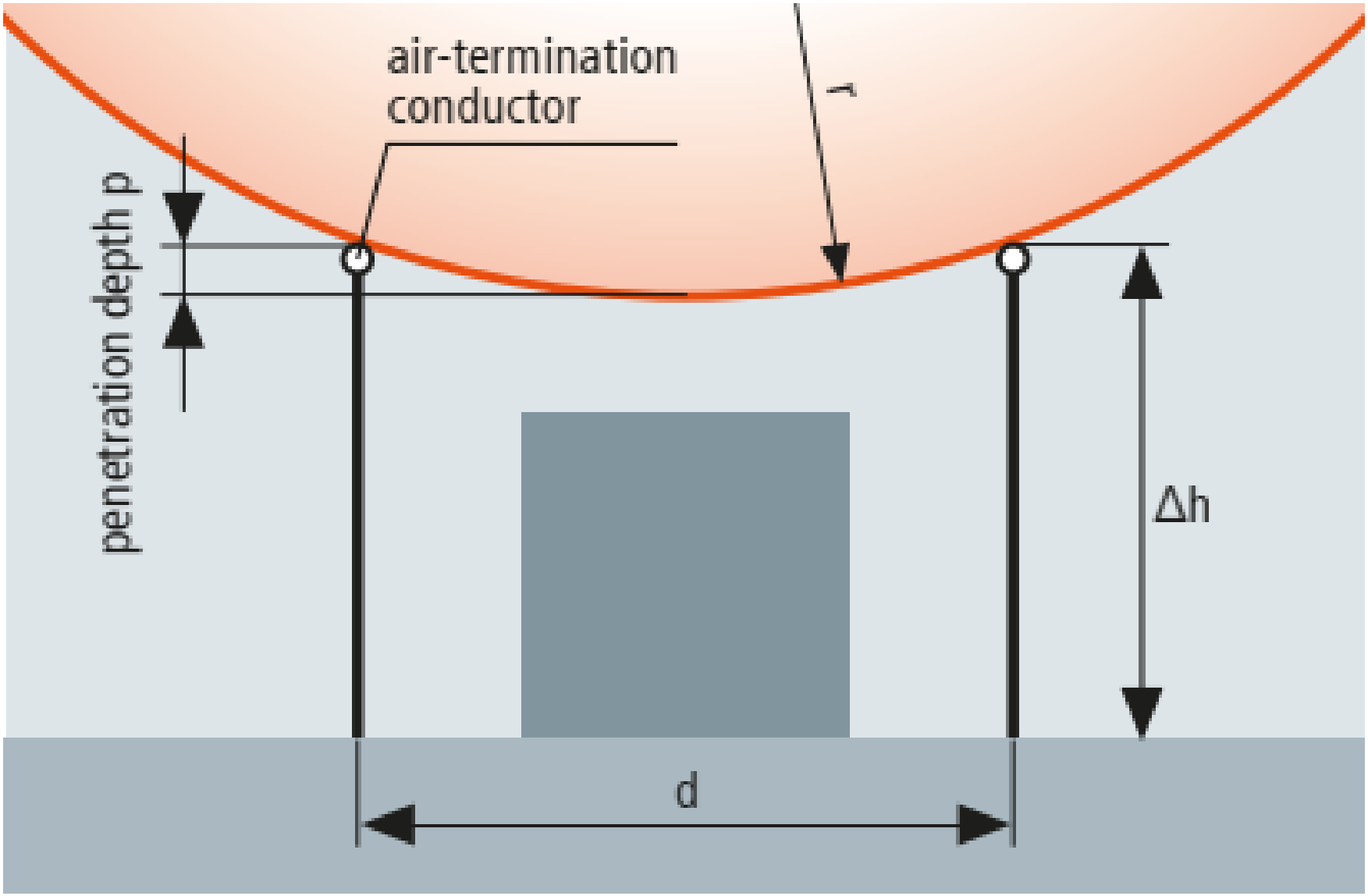


Height of the air-termination rod h in m	Class of LPS I		Class of LPS II		Class of LPS III		Class of LPS IV	
	Angle α	Distance a in m	Angle α	Distance a in m	Angle α	Distance a in m	Angle α	Distance a in m
1	71	2.90	74	3.49	77	4.33	79	5.14
2	71	5.81	74	6.97	77	8.66	79	10.29
3	66	6.74	71	8.71	74	10.46	76	12.03
4	62	7.52	68	9.90	72	12.31	74	13.95
5	59	8.32	65	10.72	70	13.74	72	15.39
6	56	8.90	62	11.28	68	14.85	71	17.43
7	53	9.29	60	12.12	66	15.72	69	18.24
8	50	9.53	58	12.80	64	16.40	68	19.80
9	48	10.00	56	13.34	62	16.93	66	20.21
10	45	10.00	54	13.76	61	18.04	65	21.45
11	43	10.26	52	14.08	59	18.31	64	22.55
12	40	10.07	50	14.30	58	19.20	62	22.57
13	38	10.16	49	14.95	57	20.02	61	23.45
14	36	10.17	47	15.01	55	19.99	60	24.25
15	34	10.12	45	15.00	54	20.65	59	24.96
16	32	10.00	44	15.45	53	21.23	58	25.61
17	30	9.81	42	15.31	51	20.99	57	26.18
18	27	9.17	40	15.10	50	21.45	56	26.69
19	25	8.86	39	15.39	49	21.86	55	27.13
20	23	8.49	37	15.07	48	22.21	54	27.53
21			36	15.26	47	22.52	53	27.87
22			35	15.40	46	22.78	52	28.16
23			36	16.71	47	24.66	53	30.52
24			32	15.00	44	23.18	50	28.60
25			30	14.43	43	23.31	49	28.76
26			29	14.41	41	22.60	49	29.91
27			27	13.76	40	22.66	48	29.99
28			26	13.66	39	22.67	47	30.03
29			25	13.52	38	22.66	46	30.03
30			23	12.73	37	22.61	45	30.00
31					36	22.52	44	29.94
32					35	22.41	44	30.90
33					35	23.11	43	30.77
34					34	22.93	42	30.61
35					33	22.73	41	30.43
36					32	22.50	40	30.21
37					31	22.23	40	31.50
38					30	21.94	39	30.77
39					29	21.62	38	30.47
40					28	21.27	37	30.14
41					27	20.89	37	30.90
42					26	20.48	36	30.51
43					25	20.05	35	30.11
44					24	19.59	35	30.81
45					23	19.10	34	30.35
46							33	29.87
47							32	29.37
48							32	29.99
49							31	29.44
50							30	28.87
51							30	29.44
52							29	28.82
53							28	28.18
54							27	27.51
55							27	28.02
56							26	27.31
57							25	26.58
58							25	27.05

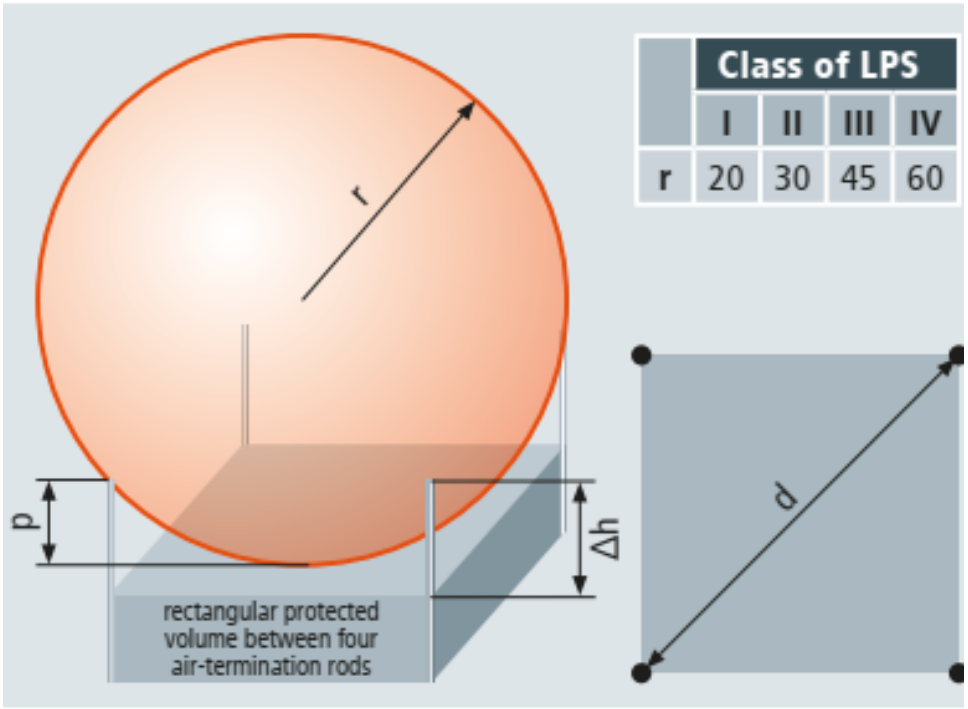


1.4. Air Terminal Design:
Rolling Sphere Methode

Penetration Depth of The Rolling Sphere

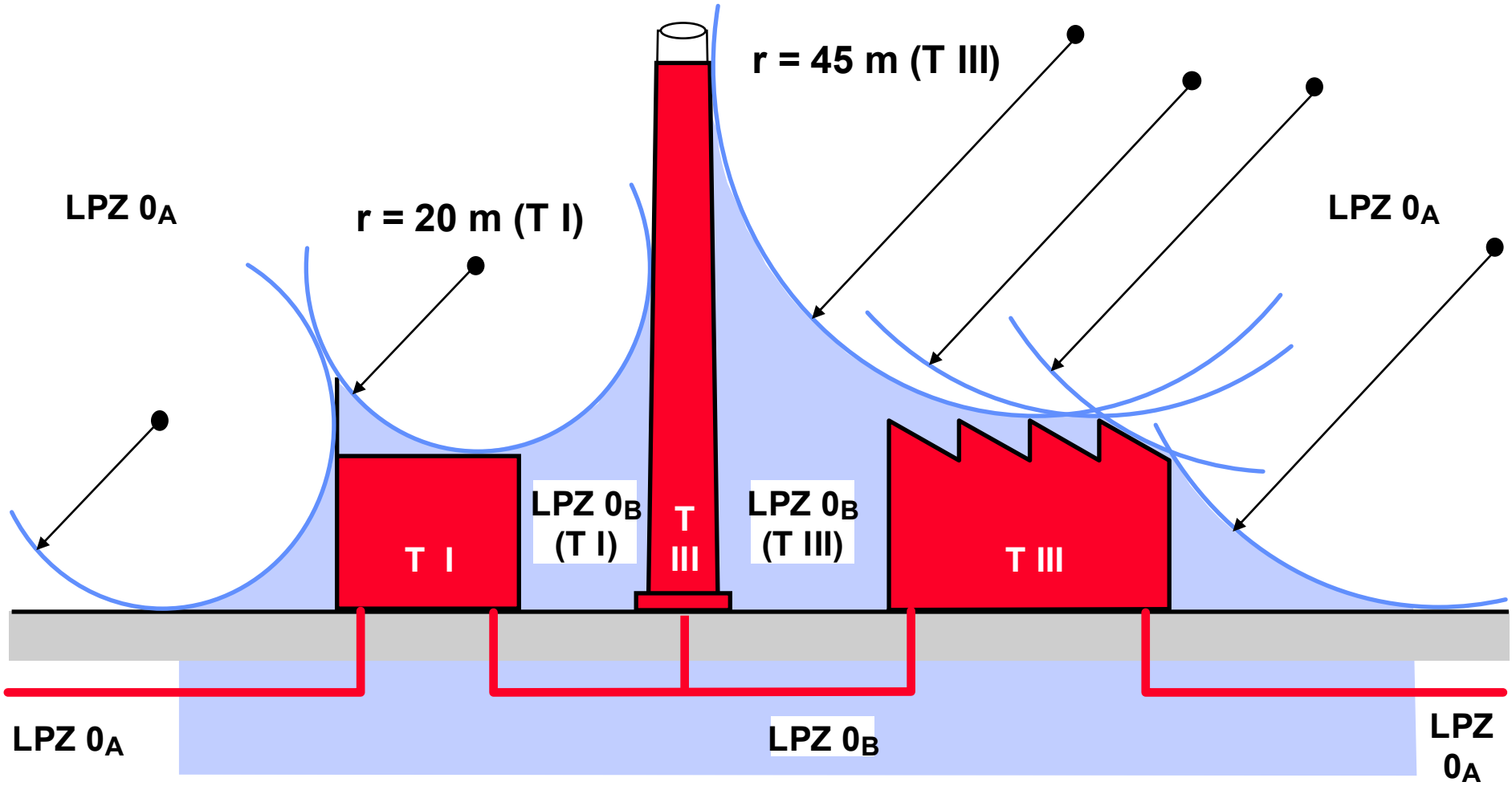


Air-Termination System for Roof-Mounted Structures and Their Protected Volume



d	Sag of the rolling sphere [m] (rounded up)			
Distance between the air-termination rods [m]	Class of LPS with rolling sphere radius [m]			
	I (20 m)	II (30 m)	III (45 m)	IV (60 m)
2	0.03	0.02	0.01	0.01
4	0.10	0.07	0.04	0.03
6	0.23	0.15	0.10	0.08
8	0.40	0.27	0.18	0.13
10	0.64	0.42	0.28	0.21
12	0.92	0.61	0.40	0.30
14	1.27	0.83	0.55	0.41
16	1.67	1.09	0.72	0.54
18	2.14	1.38	0.91	0.68
20	2.68	1.72	1.13	0.84
23	3.64	2.29	1.49	1.11
26	4.80	2.96	1.92	1.43
29	6.23	3.74	2.40	1.78
32	8.00	4.62	2.94	2.17
35	10.32	5.63	3.54	2.61

Lightning Protection Zones (LPZ)
LPS Type (T)



LPZ: Lightning Protection Zone
T : Lightning Protection System Type

1.5. GROUNDING

Sistem Pentanahan

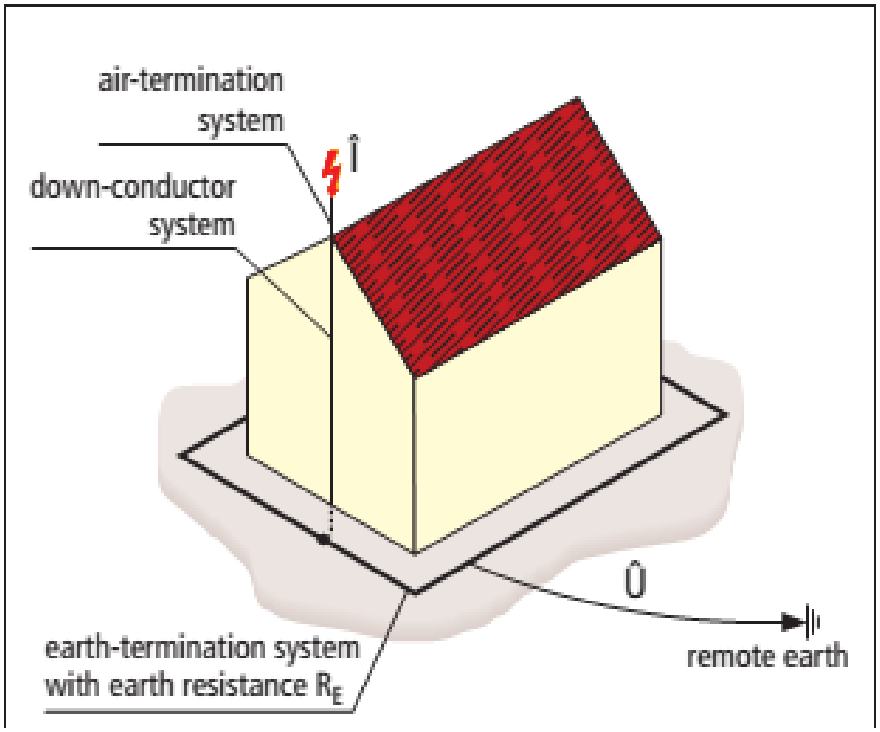
Sistem Pentanahan adalah sistem yang menghubungkan elektroda pentanahan yang konduktif atau komponen metal dari struktur yang terhubung dengan bumi.

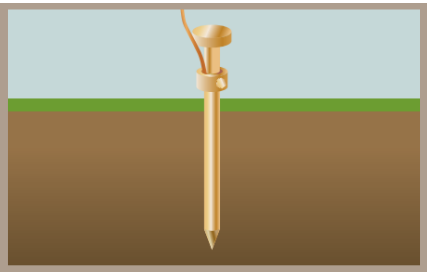
Elektroda pentanahan bisa berupa material silinder atau plat dan dirancang dengan konfigurasi star, ring atau meshed atau kombinasinya.

Elektroda pentanahan memiliki tiga tipe, yakni:

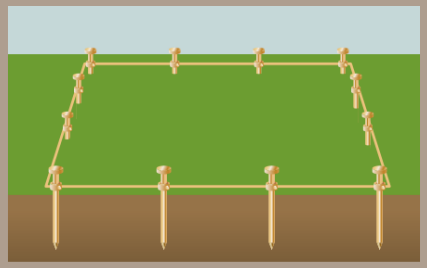
- Natural, bagian metal bangunan yang awalnya dirancang bukan sebagai elektroda
- Pondasi, konduktor yang di dalam pondasi beton
- Kendali, berfungsi lebih pada kendali tegangan daripada resistensi.

Earth Termination System

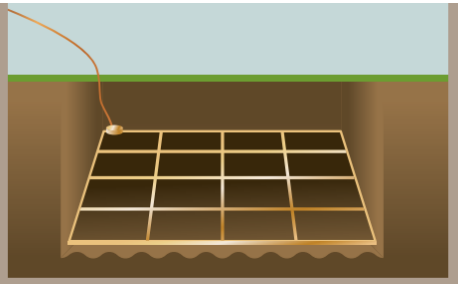




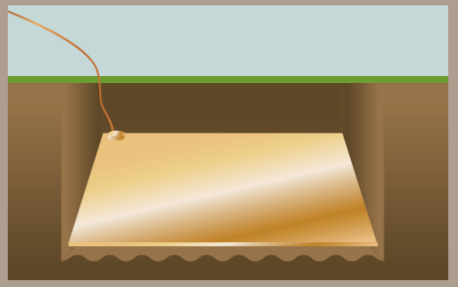
Single ground electrode.



Multiple ground electrodes connected.



Mesh network.



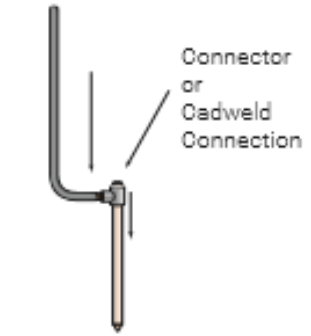
Ground plate.

Type A – Vertical & Horizontal Electrodes

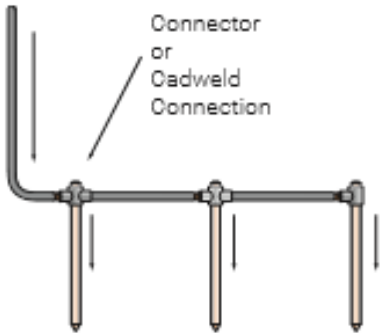
Vertical and/or horizontal electrodes are used. There must be earth electrodes installed at the base of each down-conductor. A minimum of two electrodes must be used.

The minimum total length of electrode at each down-conductor is:

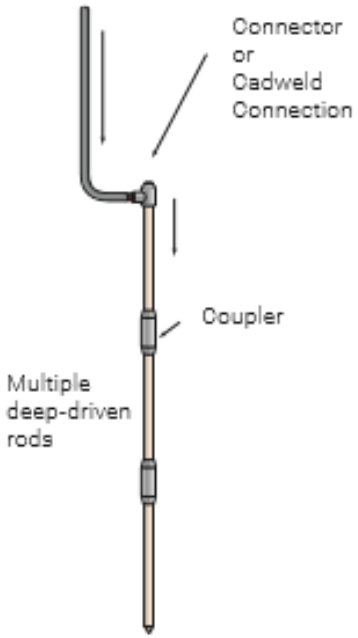
- l_1 for horizontal electrodes
- 0.5 l_1 for vertical (or inclined) electrodes



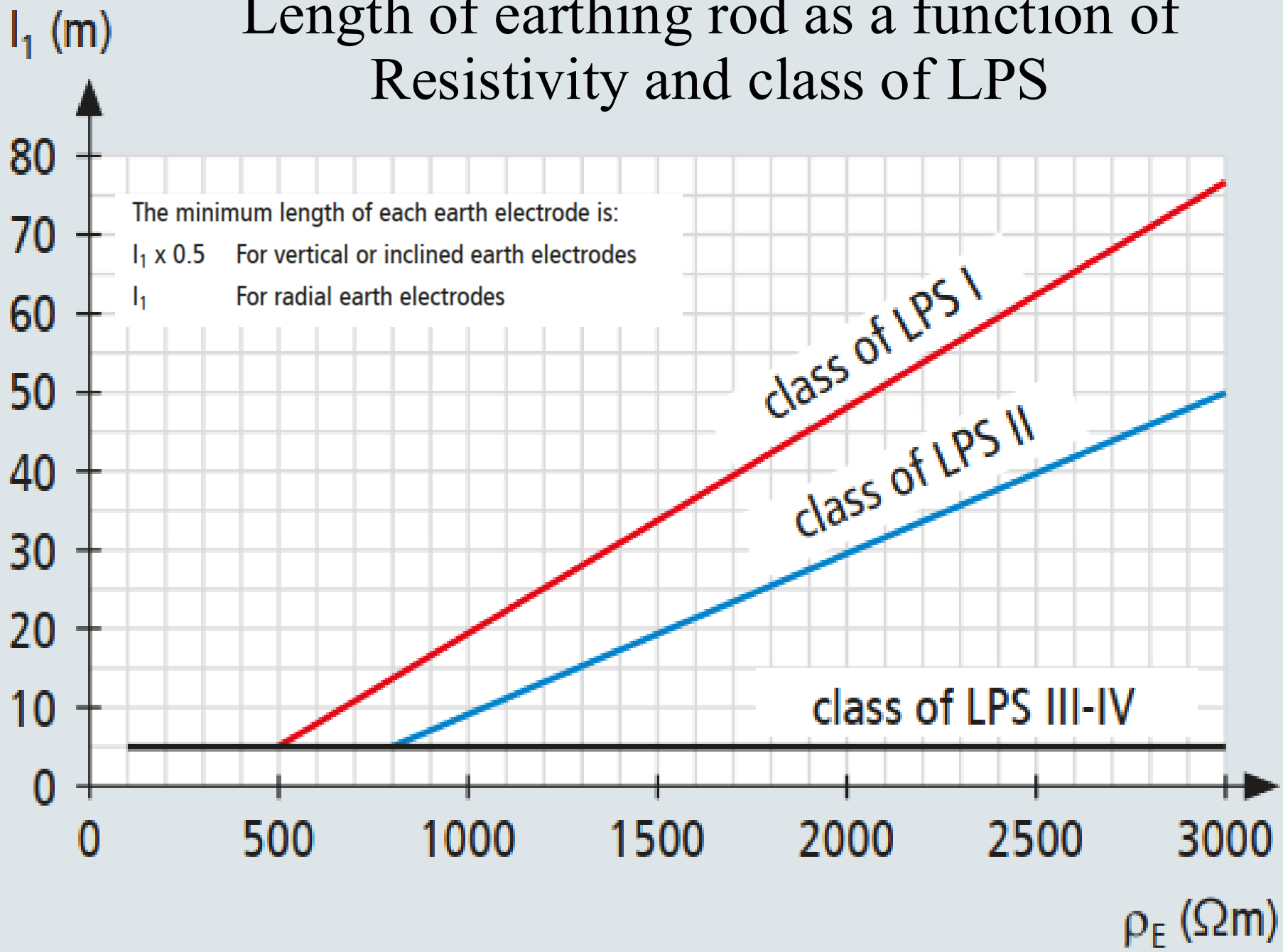
Single ground rod



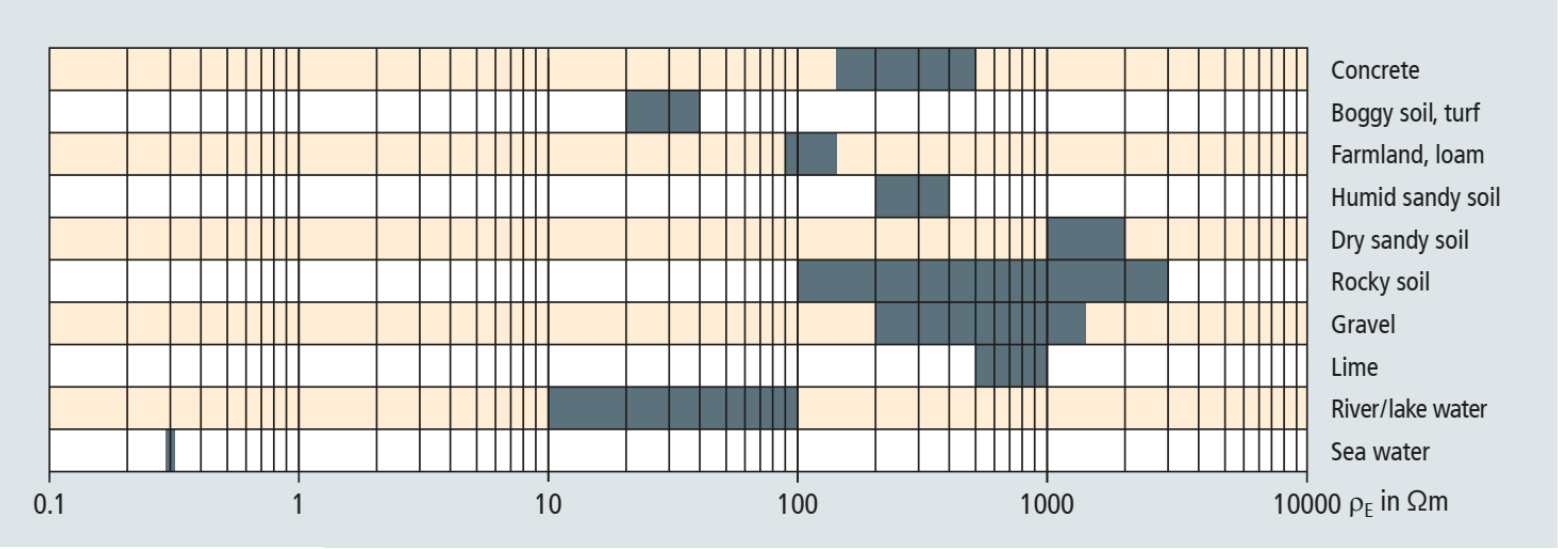
Multiple ground rods in a line, ring, or crows foot arrangement.



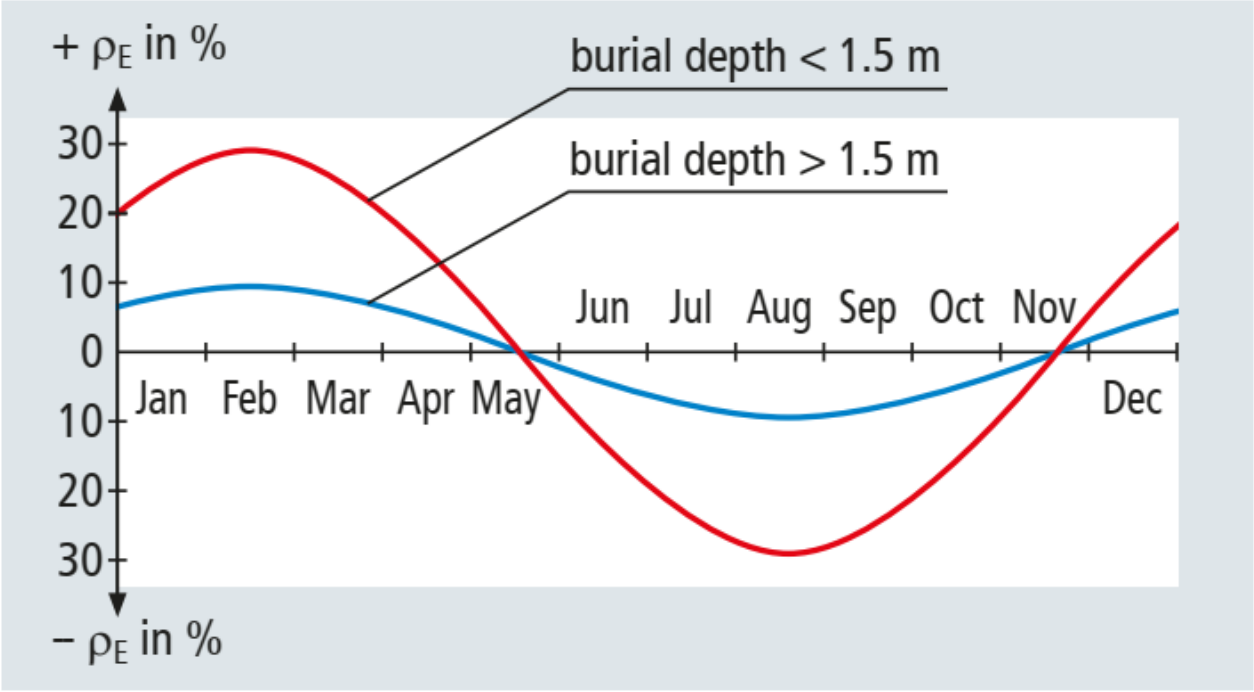
Length of earthing rod as a function of Resistivity and class of LPS

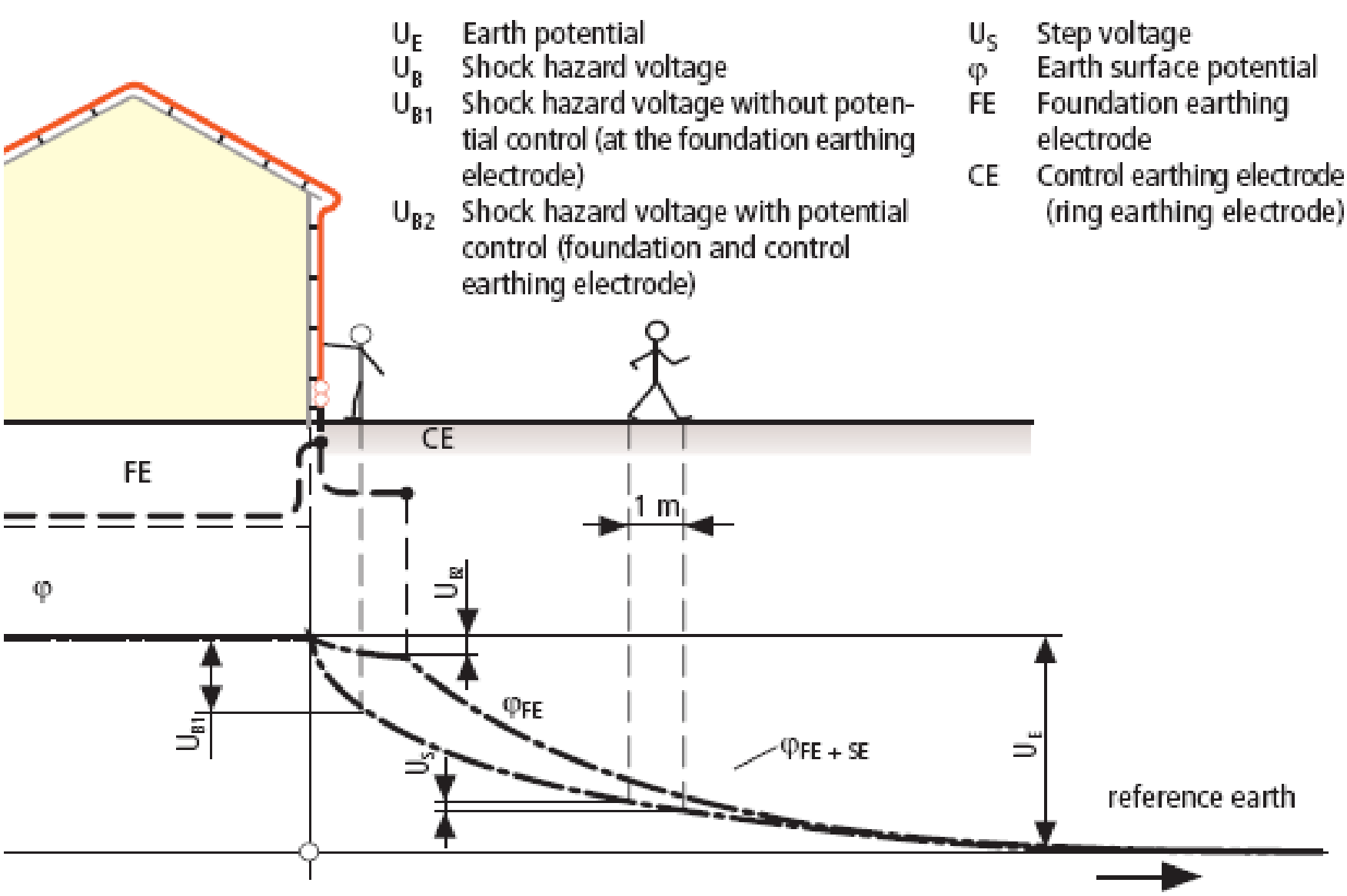


Soil Resistivity



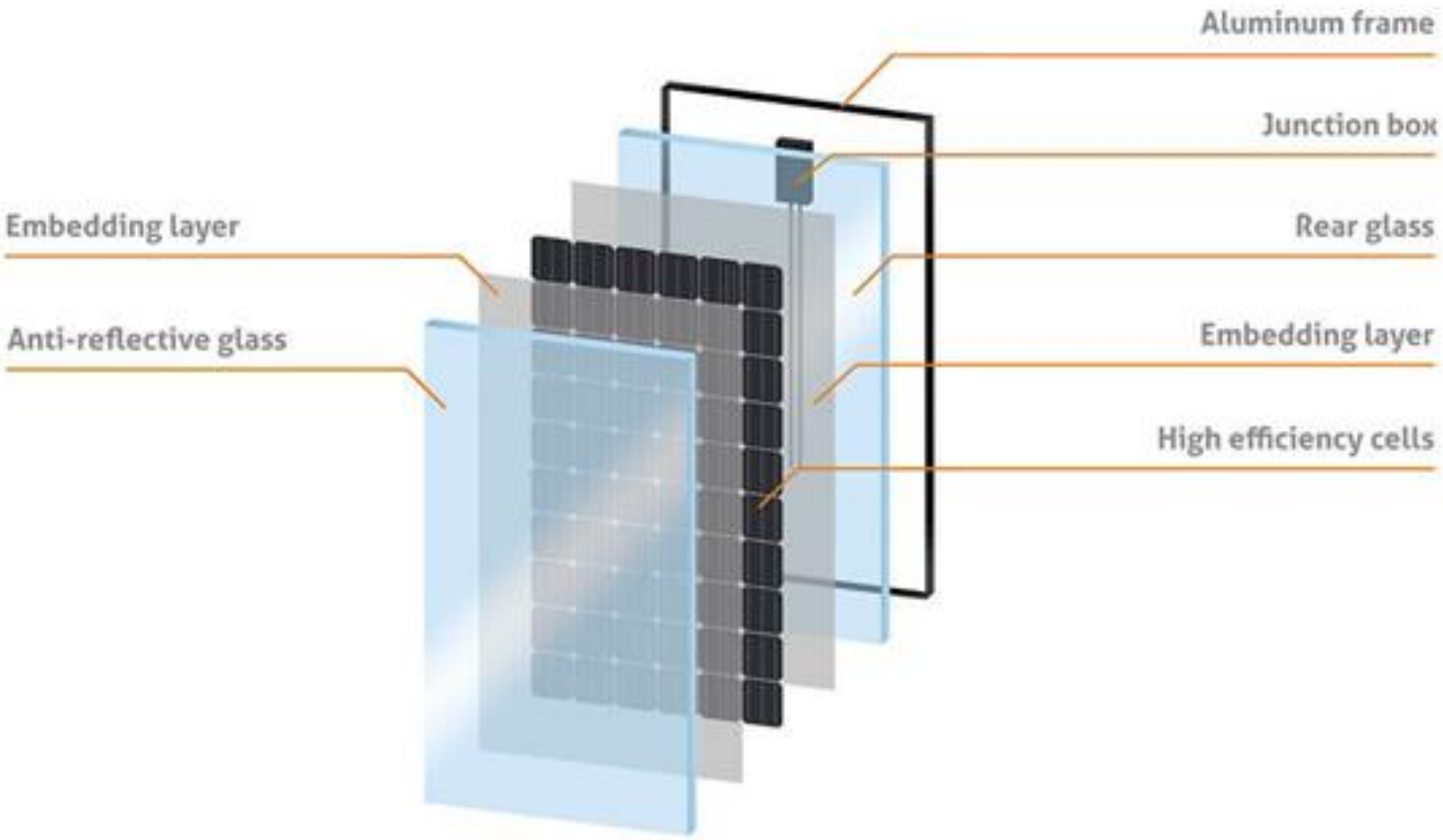
Type of soil	Soil resistivity R_E
	ΩM
Very moist soil, swamplike	30
Farming soil loamy and clay soils	100
Sandy clay soil	150
Moist sandy soil	300
Concrete 1:5	400
Moist gravel	500
Dry sandy soil	1000
Dry gravel	1000
Stoney soil	30,000
Rock	10^7




U_E Earth potential
U_B Shock hazard voltage
U_{B1} Shock hazard voltage without potential control (at the foundation earthing electrode)
U_{B2} Shock hazard voltage with potential control (foundation and control earthing electrode)
U_S Step voltage
φ Earth surface potential
FE Foundation earthing electrode
CE Control earthing electrode (ring earthing electrode)
CE
FE
1 m
φ
U_{B2}
U_{B1}
U_S
φ_{FE}
φ_{FE+SE}
U_E
reference earth

III. EXTERNAL INTERNAL LIGHTNING PROTECTION FOR PV - PLANT

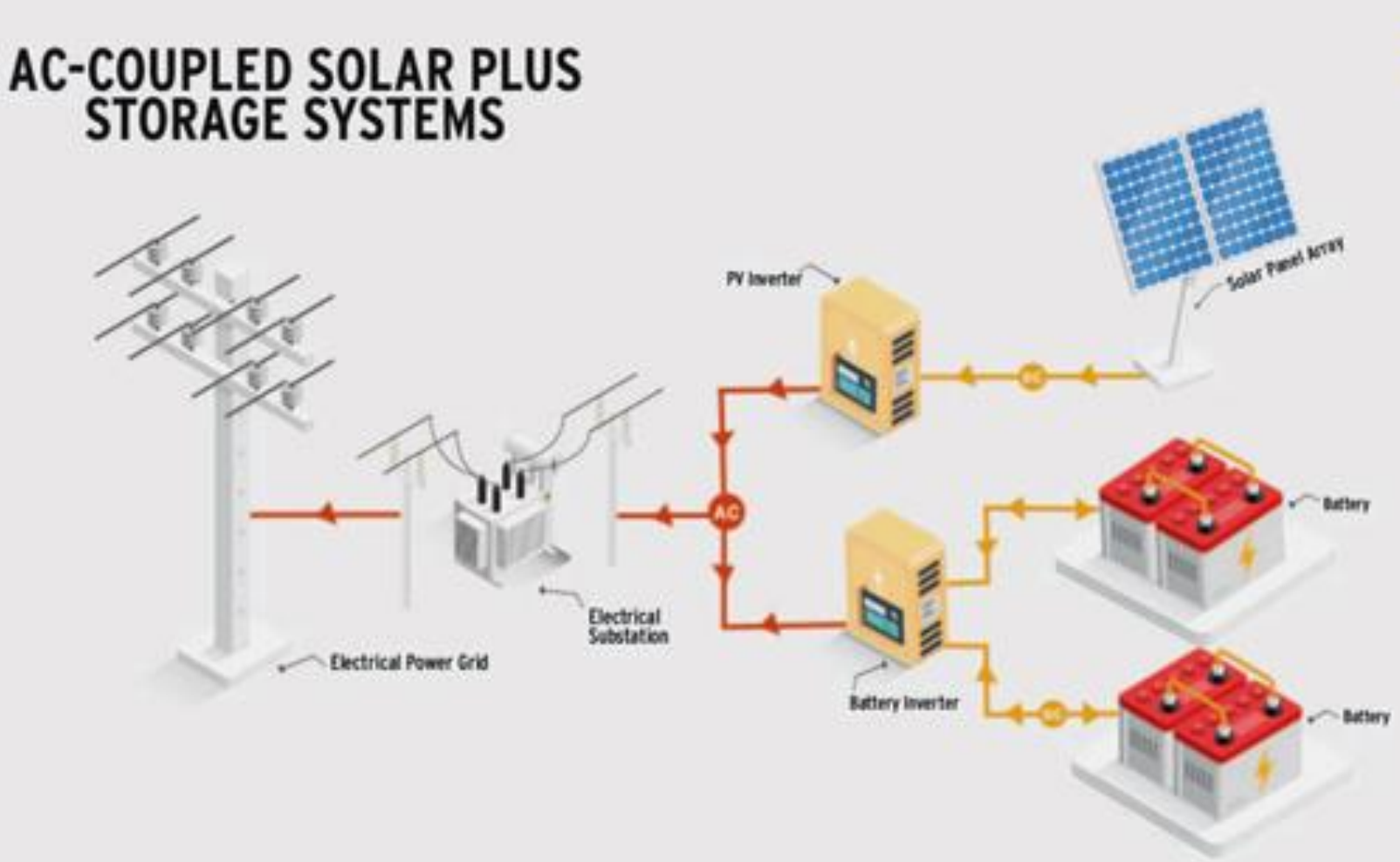
Lapisan Modul Solar PV



Monocrystalline Module

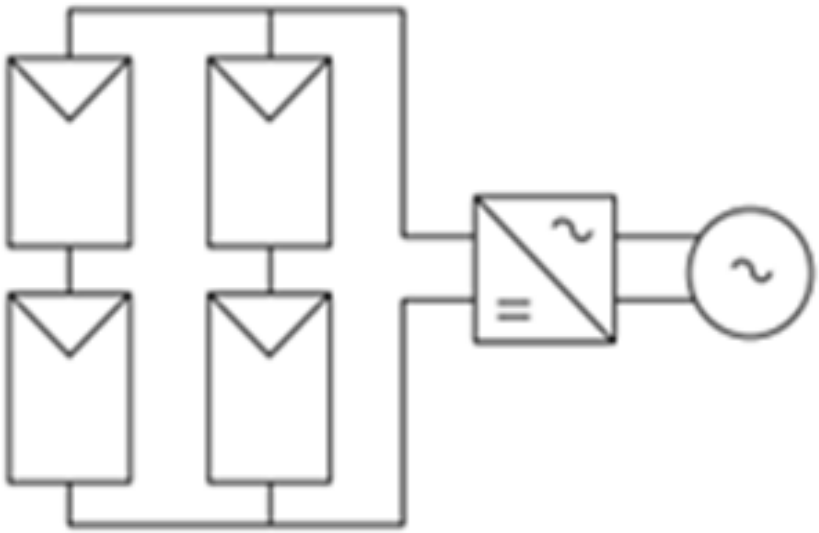


Diagram Sistem Pembangkit Listrik Tenaga Surya yang dilengkapi Battery

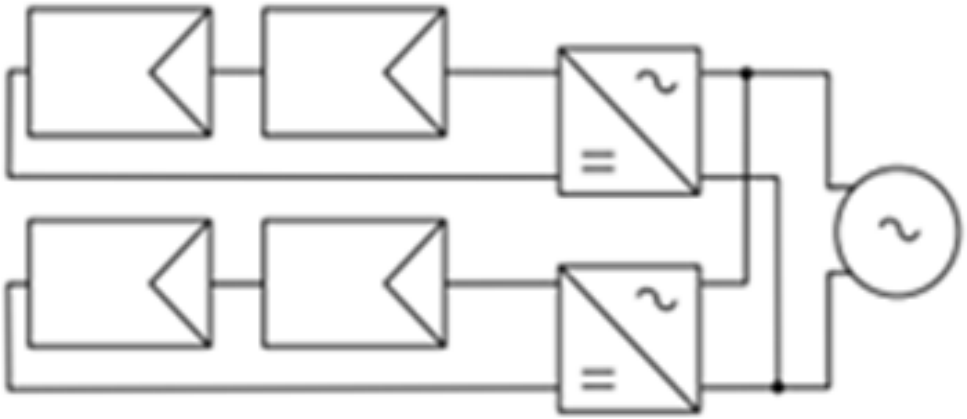


Konfigurasi Sistem Inverter

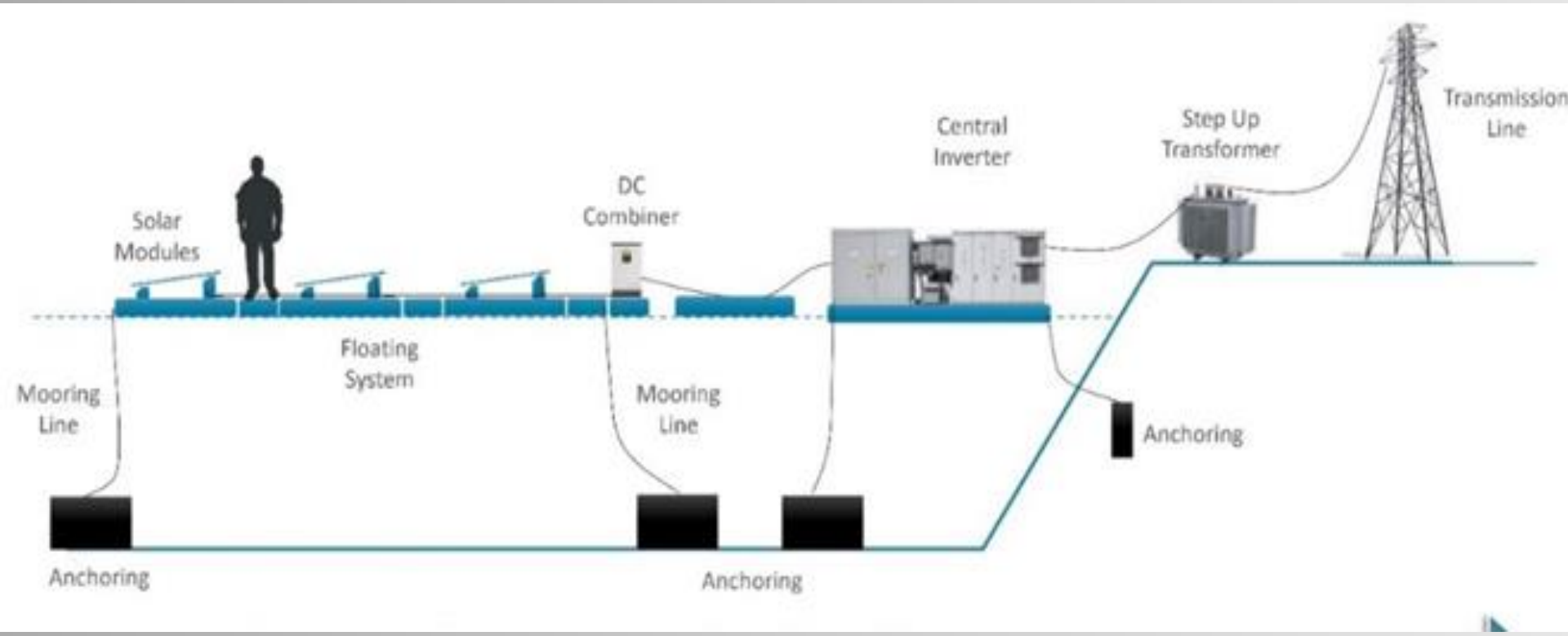
Central Inverter



String Inverter

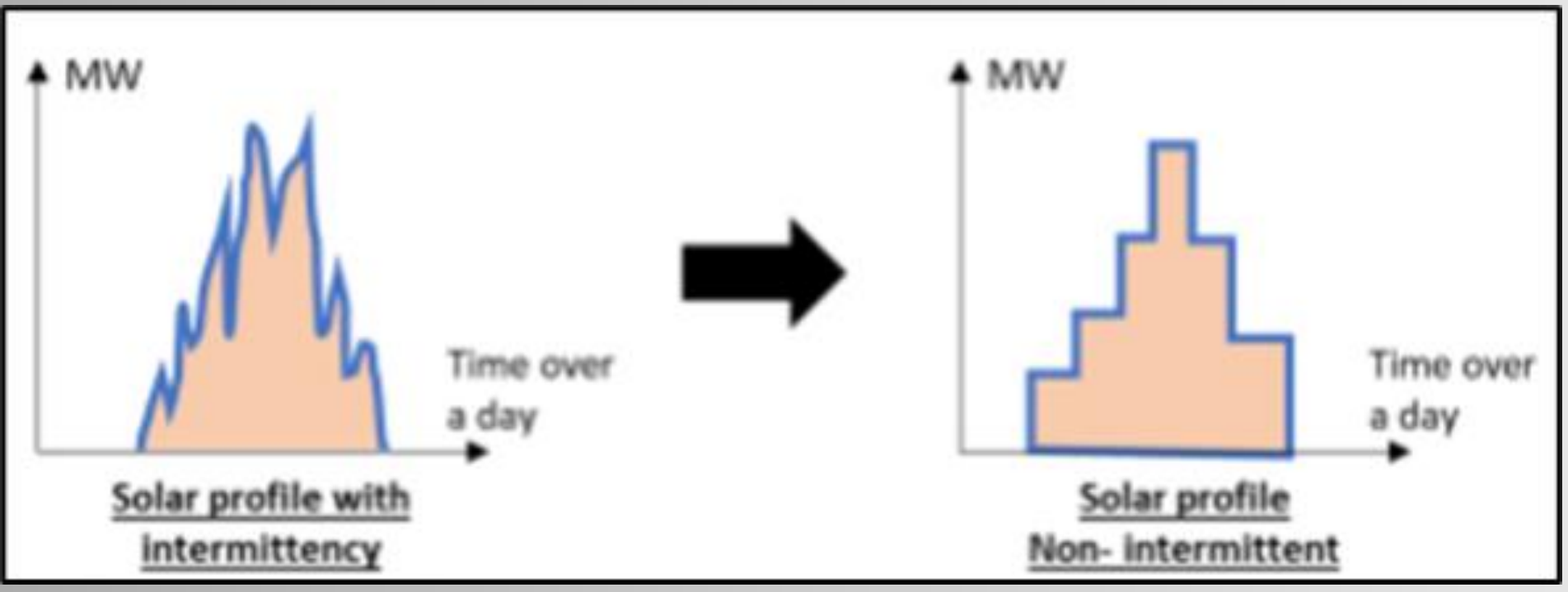


Skematik PLTS Terapung



Ilustrasi Energi Intermiten dan Non-Intermiten

Energi listrik yang dihasilkan dari teknologi PV merupakan energi listrik intermittent, energi listrik tidak konstan sepanjang hari, berfluktuasi sesuai dengan ketersediaan sinar matahari berdampak ketidakstabilan pada jaringan P



- Perlu pengadaan, pemasangan Battery Energy Storage System (BESS) merupakan penyimpanan energi yang bertujuan menghasilkan energi yang konstan dan stabil.

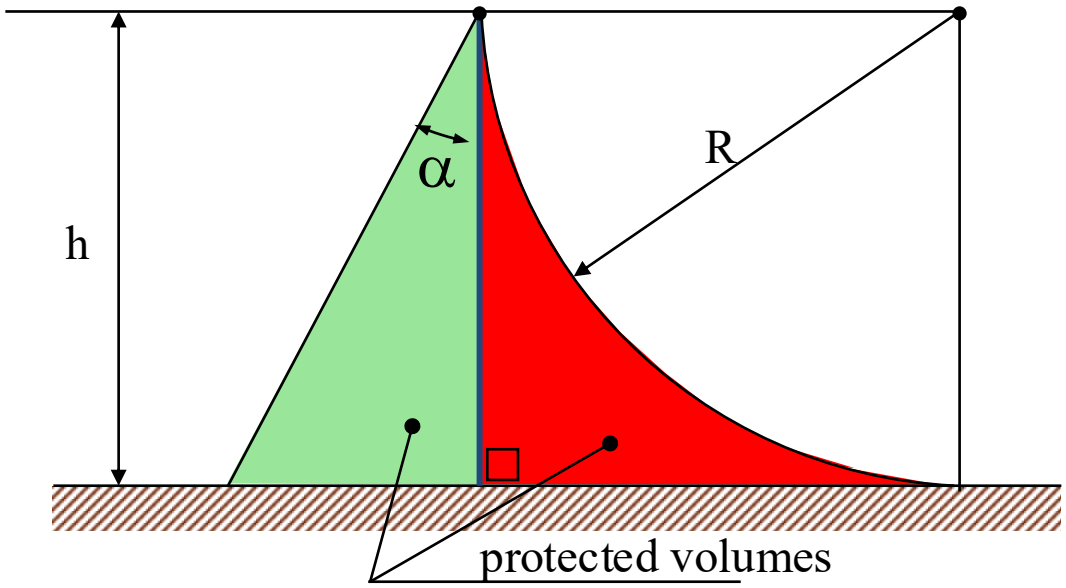
Masalah

- Premature aging akibat induksi Elektrostatis awan bermuatan,
- Prematur aging akibat induksi elektromagnetik arus sambaran petir
- Sambaran Petir langsung pada PV Panel
- Kerusakan komponen elektroteknik akibat propagasi tegangan lebih petir

Proposed solution

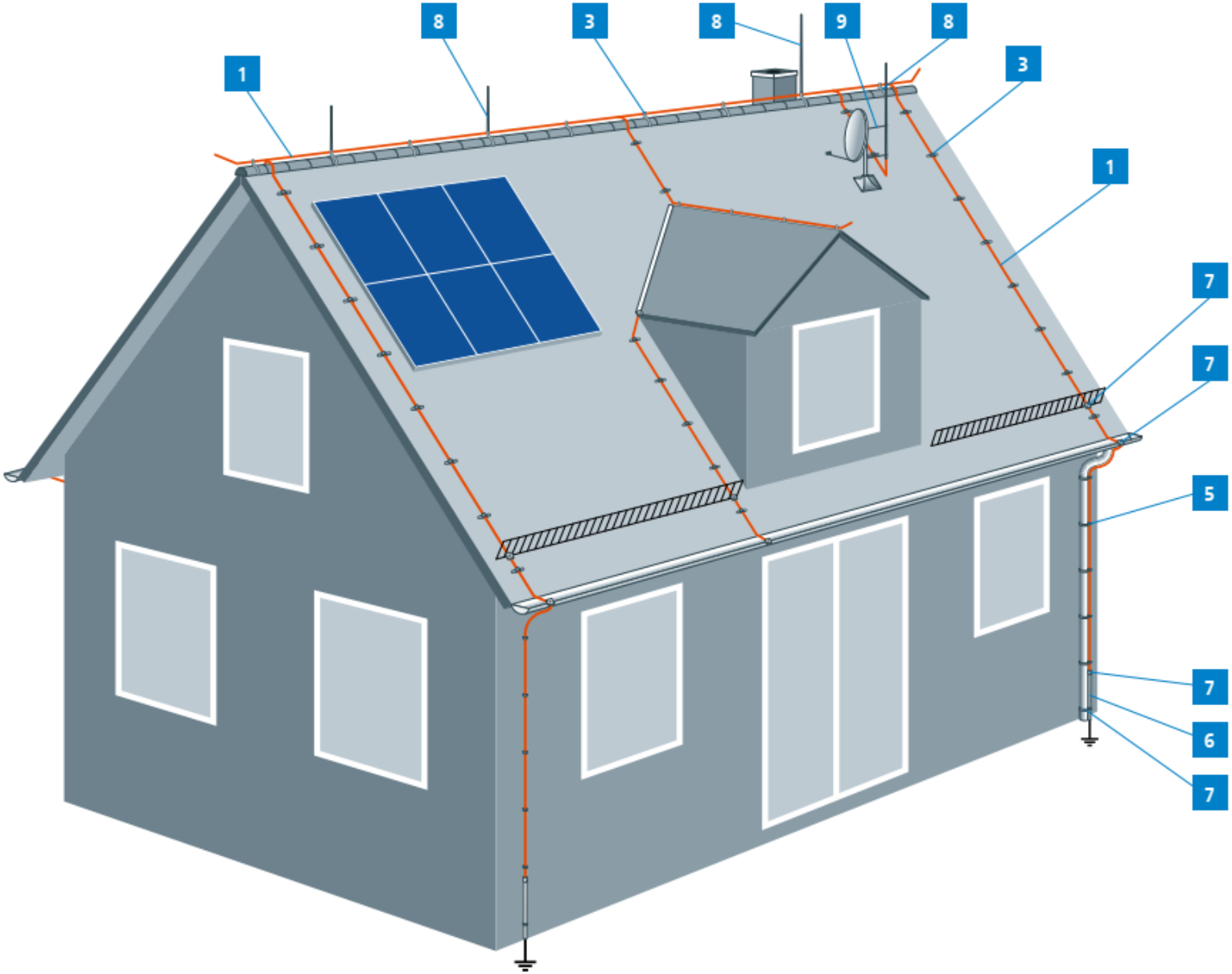
- **Penyediaan direct grounding electrode untuk pengaliran muatan induksi elektrostatis pada PV Frame dan hindari terjadinya impedance loop.**
- **Pembatasan sambaran petir pada PV Air terminal kurang dari 5 kA dengan pengaturan pemasangan Free Standing Air Terminal.**
- **Tidak diperkenankan menggunakan Free Standing Air Terminal jenis Non Standar Air Terminal, seperti ESE.**
- **Pemasangan Finial Air Terminal pada PV Panel.**
- **Pengadaan Internal Lightning Protection dengan pengadaan Grounding secara benar dan pemasangan Voltage Arrester**

3.1. ROOF TOP PV

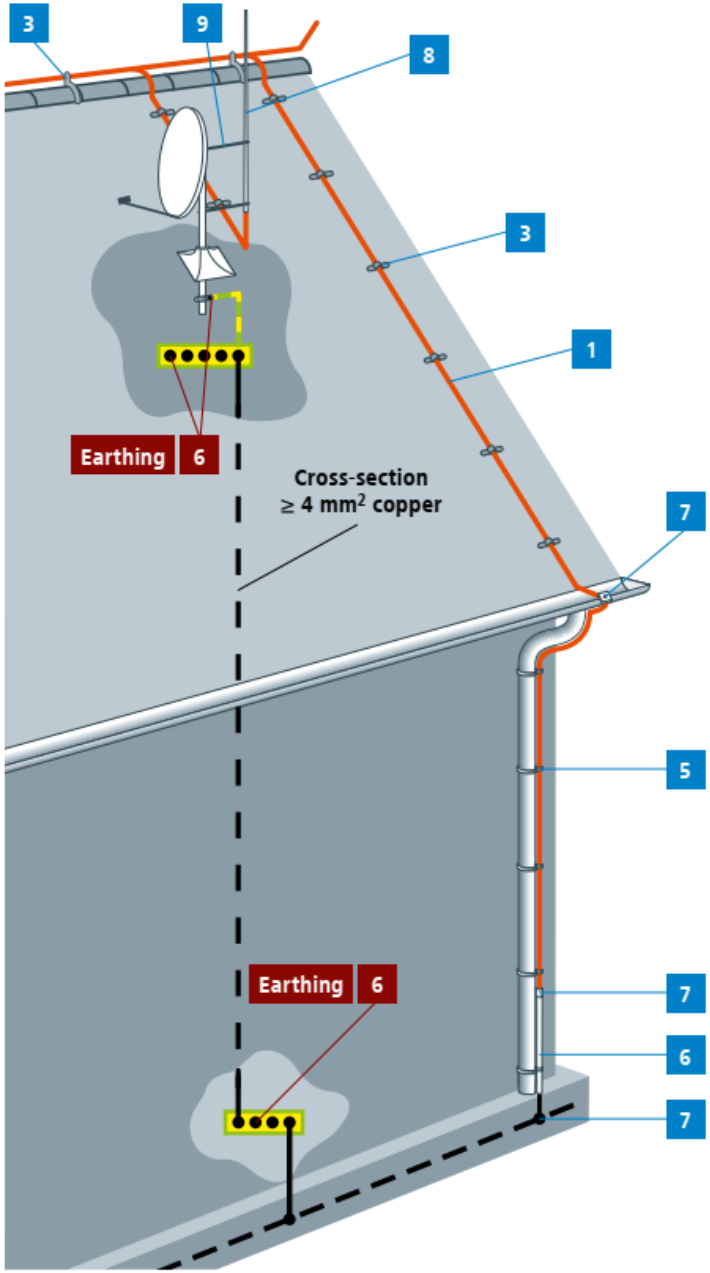


Level	R(m)	α(h = 20)	α(h = 30)	α(h = 45)	α(h = 60)	d(m)
I	20	25	*	*	*	5
II	30	35	25	*	*	10
III	45	45	35	25	*	15
IV	60	55	45	35	25	20

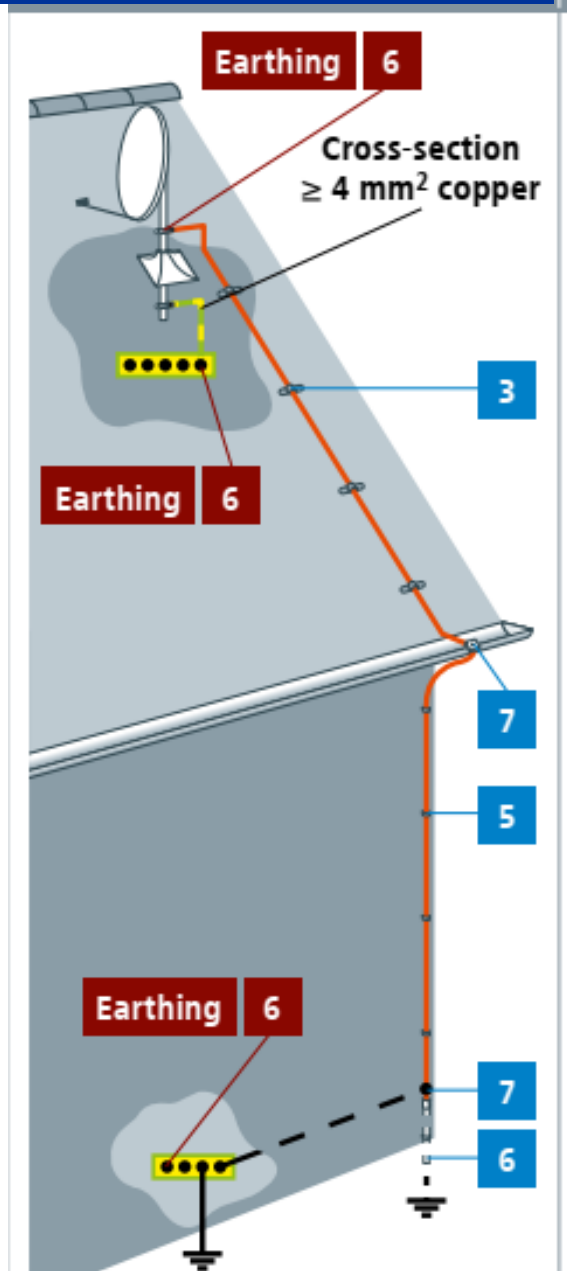
Penangkal petir atap runcing



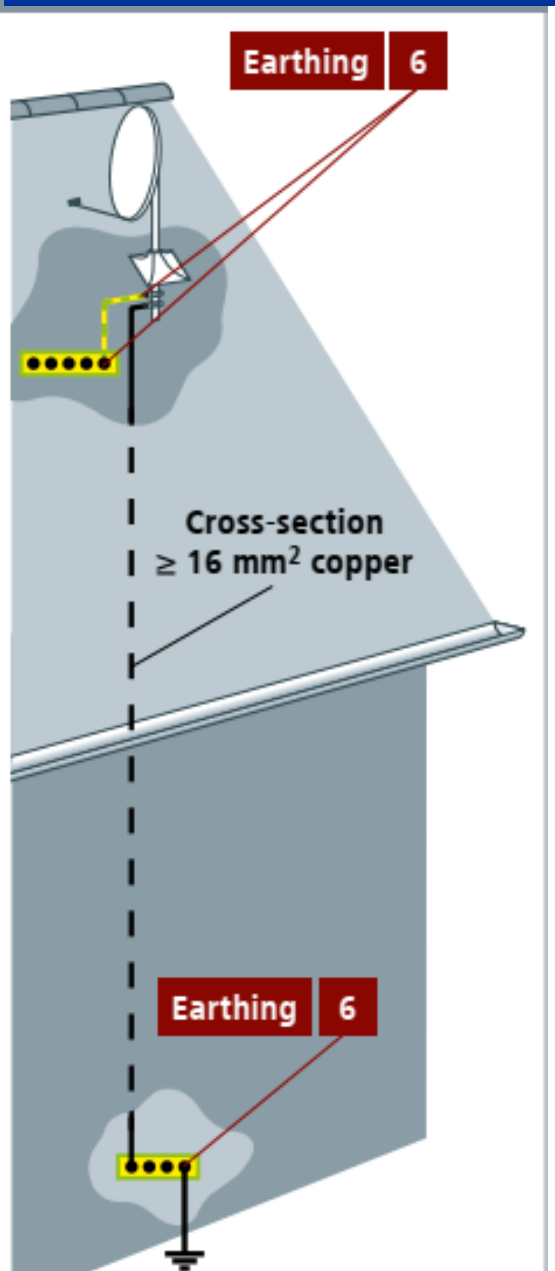
Protection of an antenna system in case of an existing lightning protection syatem according with IEC 60720-11



Equipotential bonding outside of a building without lightning protection system



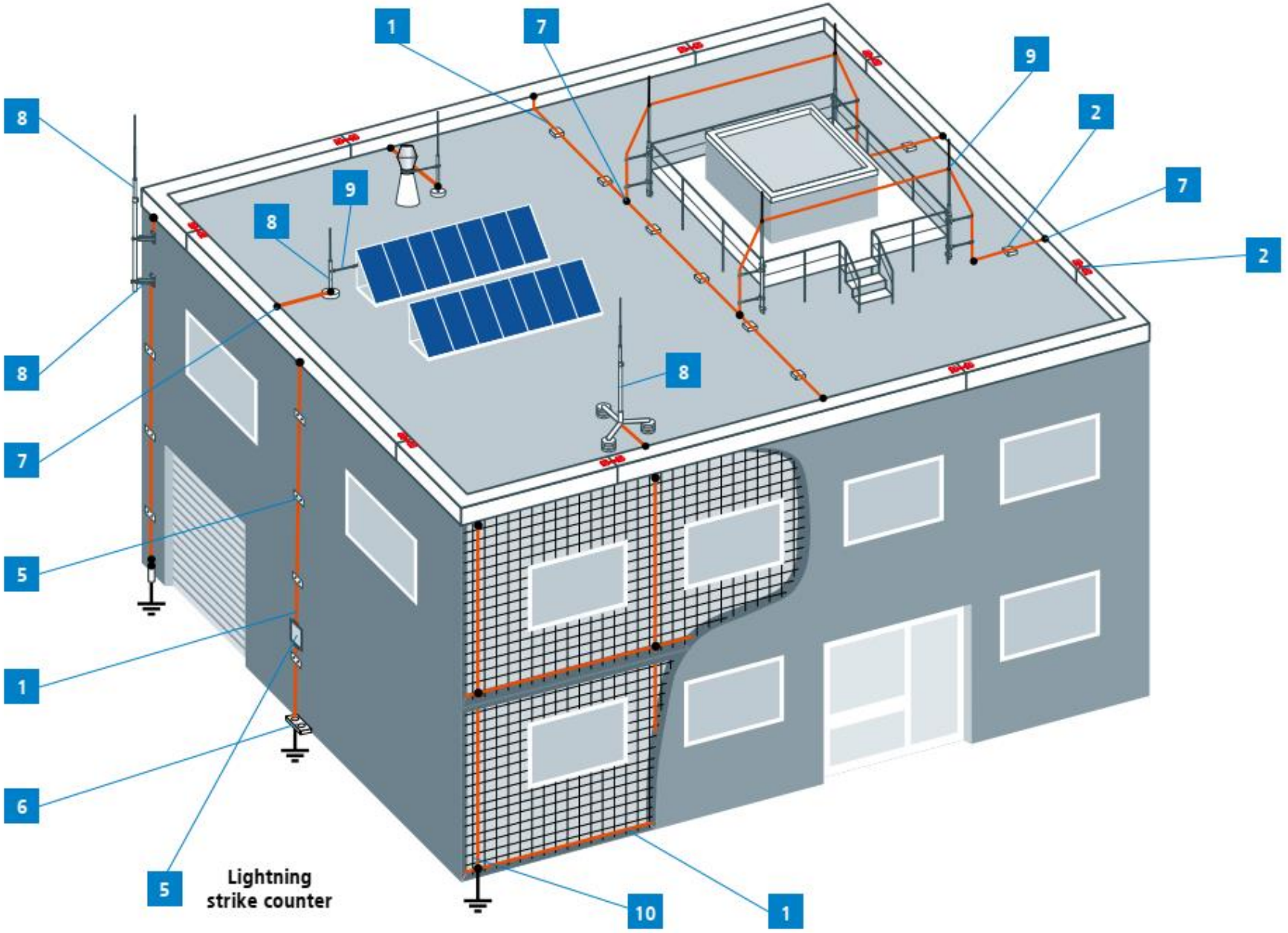
Equipotential bonding inside of a building without lightning protection system



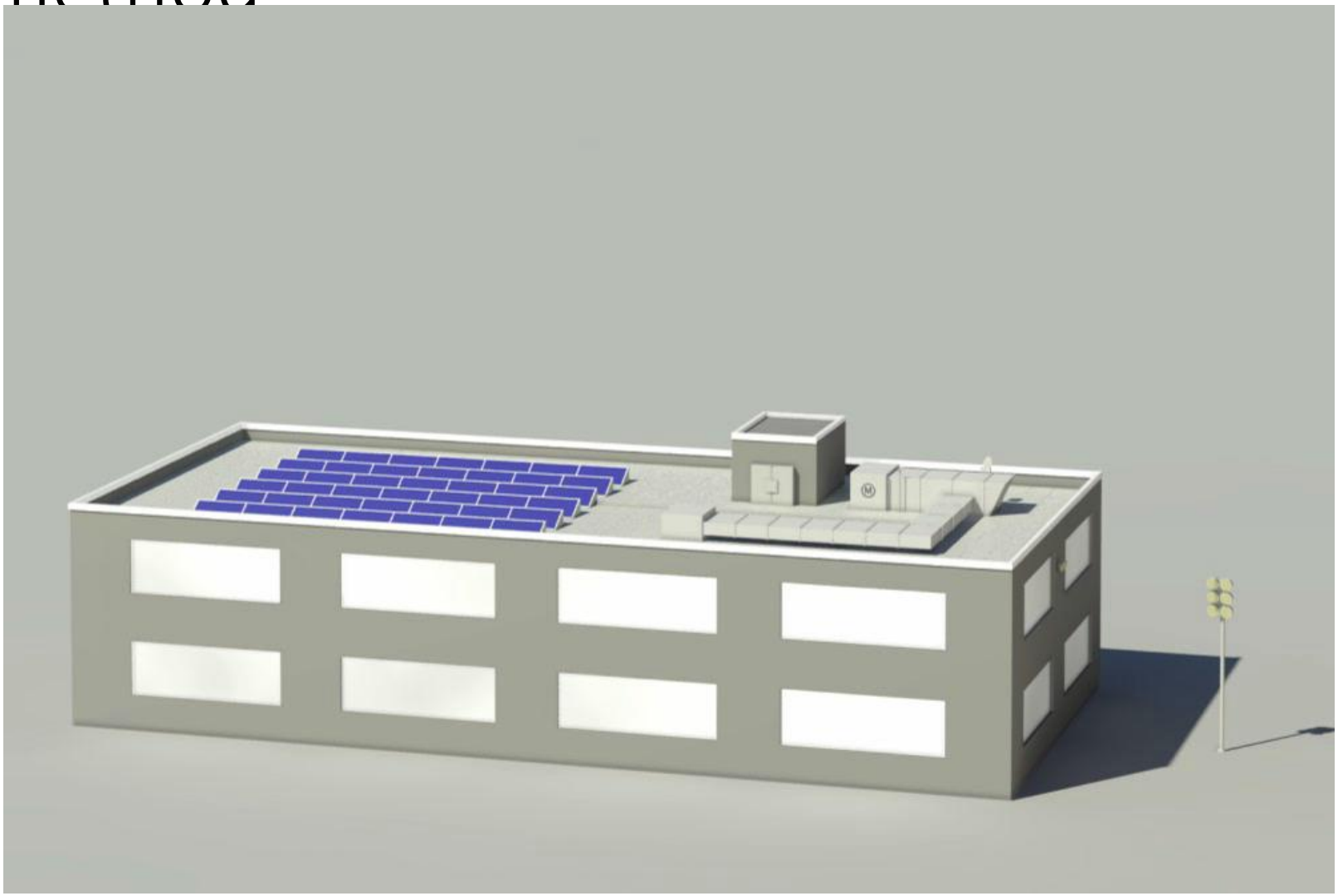
Penangkal petir atap PLTS



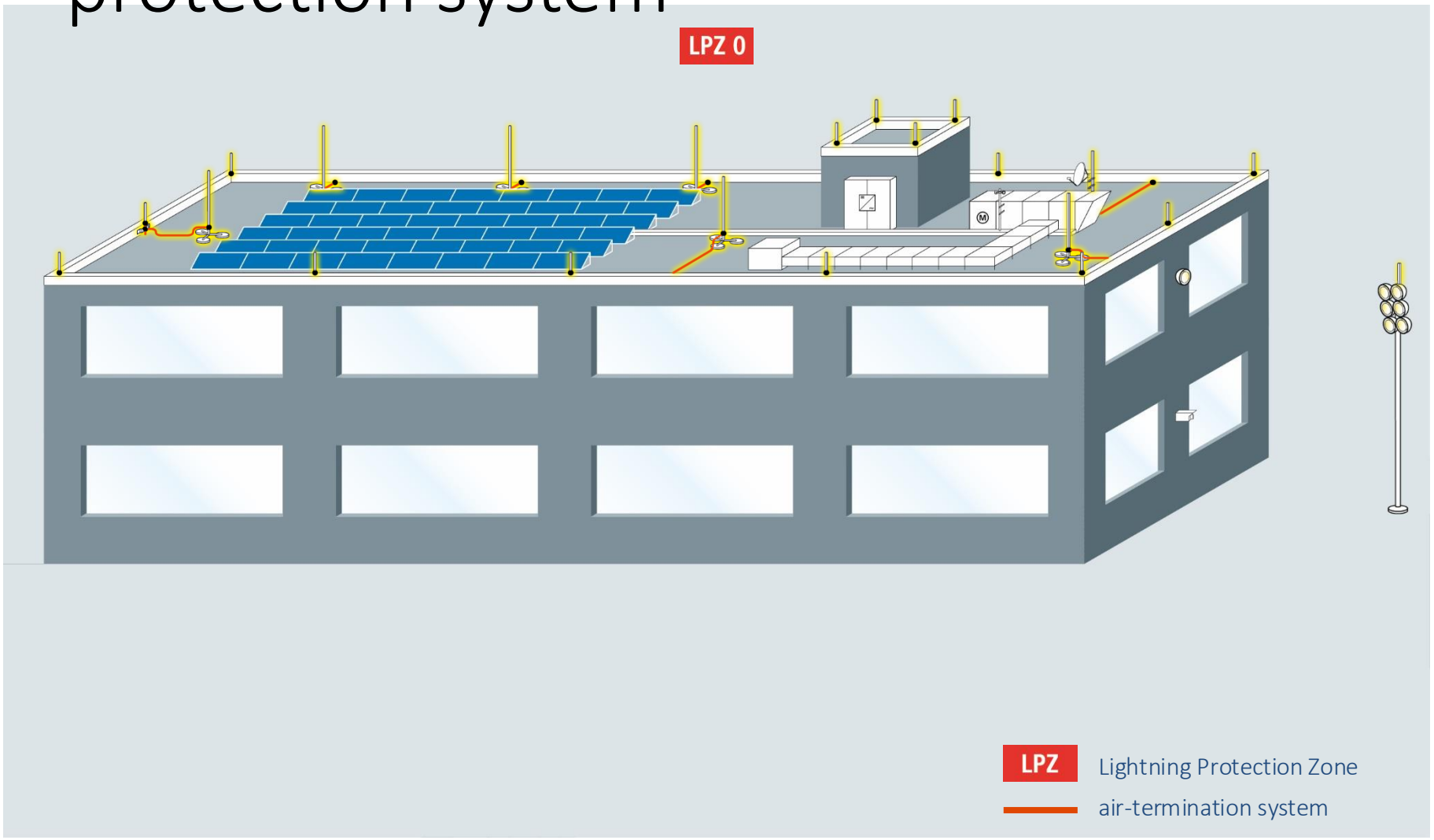
Penangkal petir atap datar



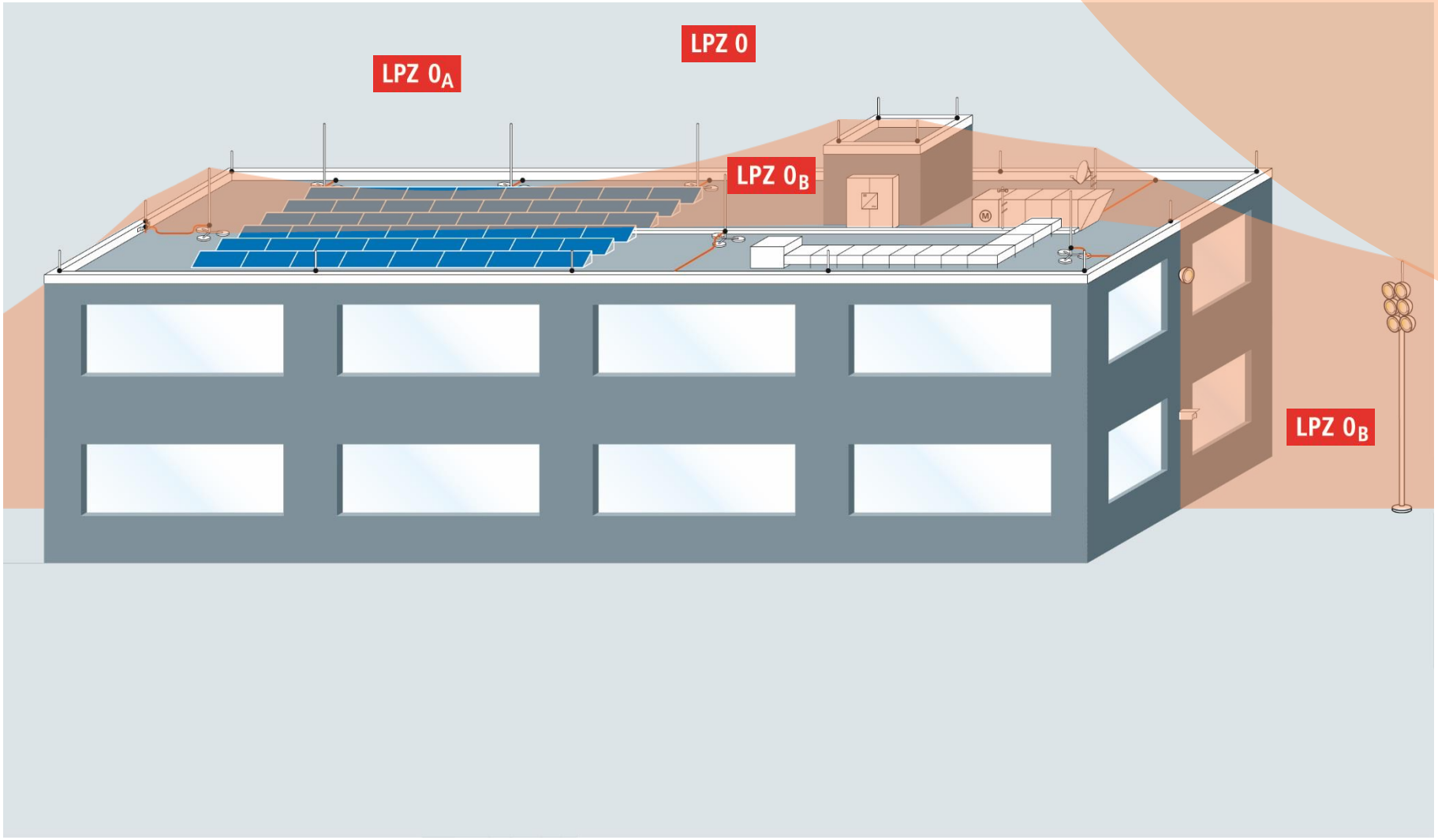
Lightning protection zones concept
Threat potential – Rolling sphere method



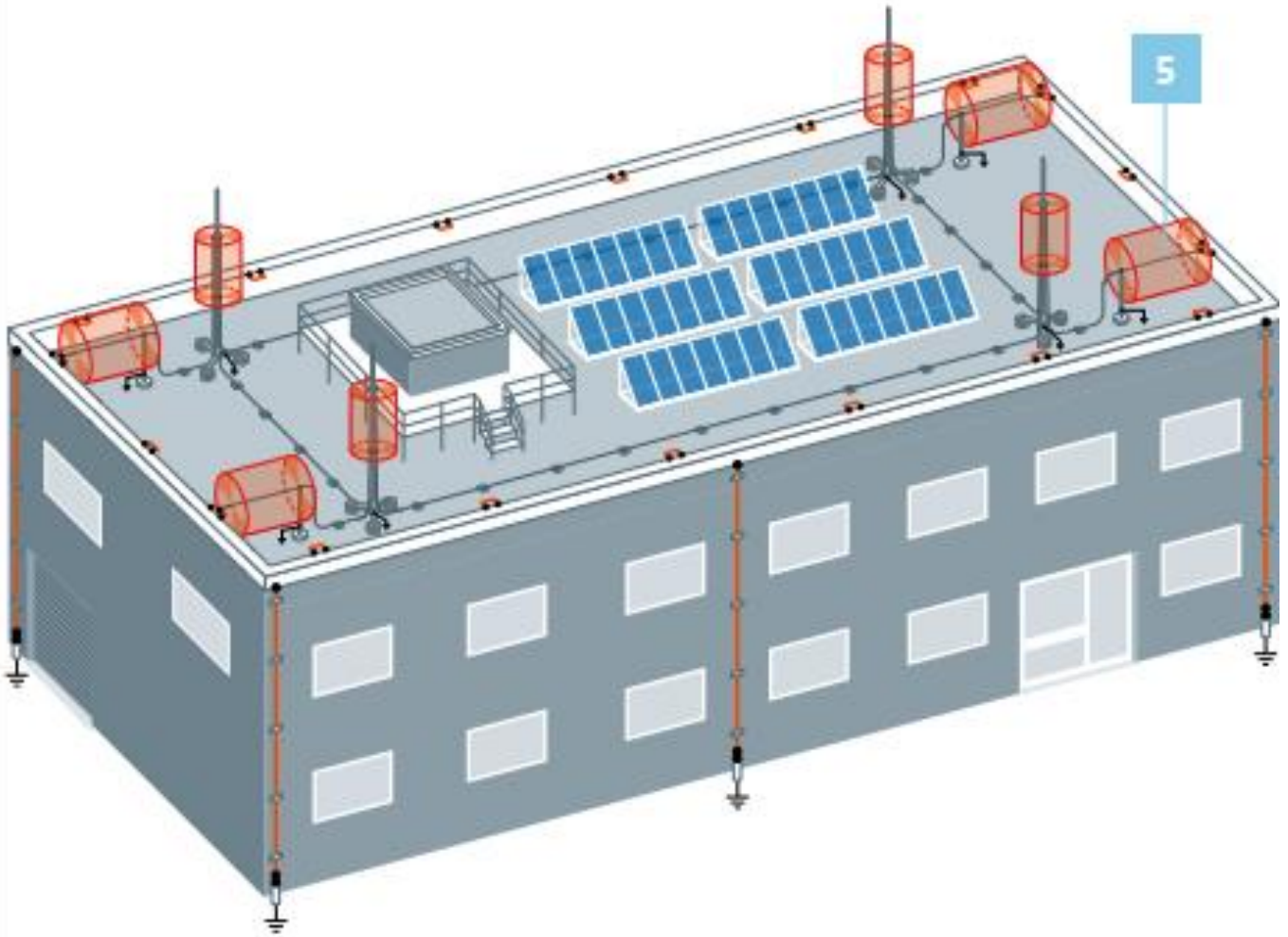
Lightning protection zone concept Building with external lightning protection system



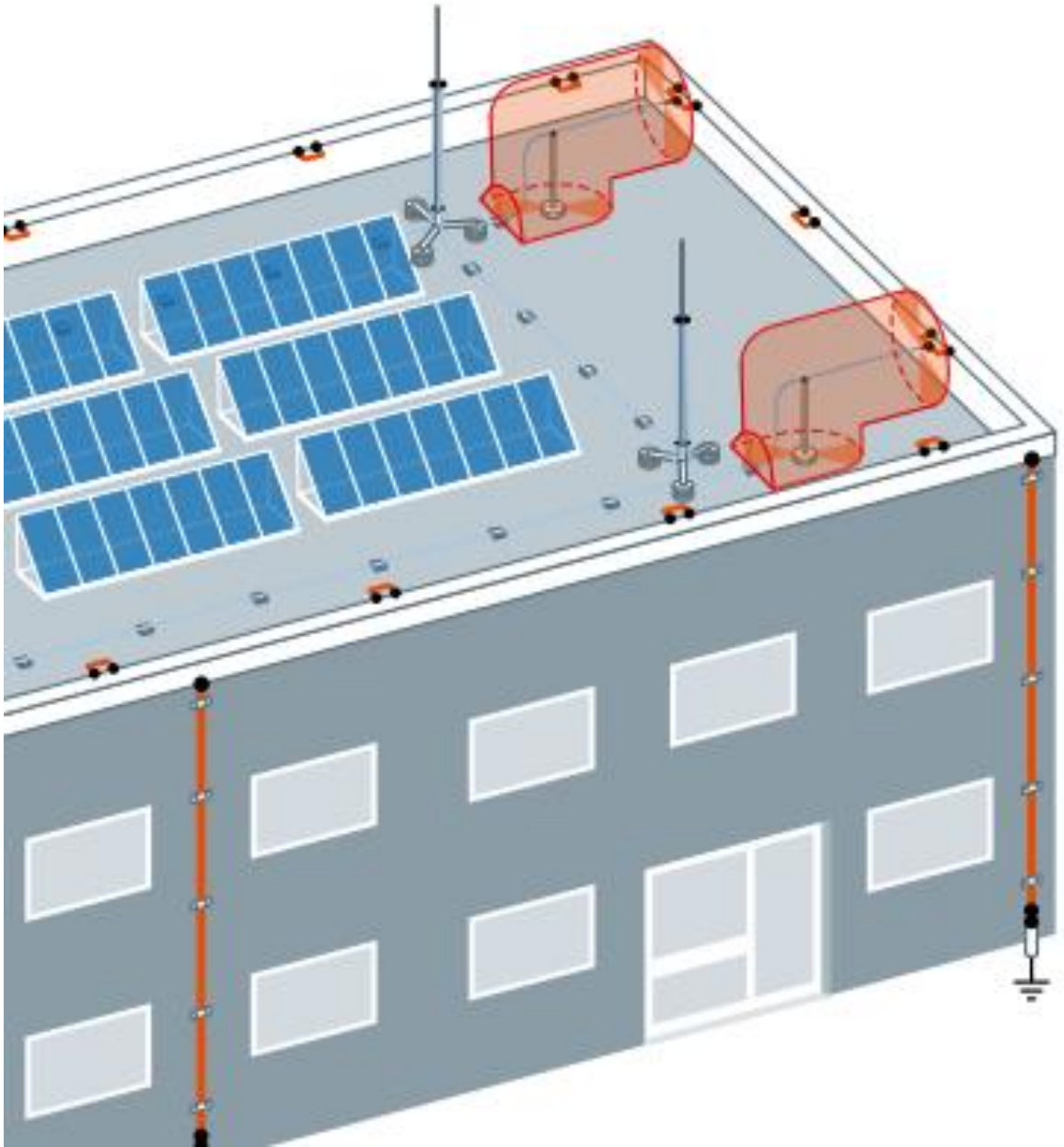
Lightning protection zone concept
Rolling sphere method



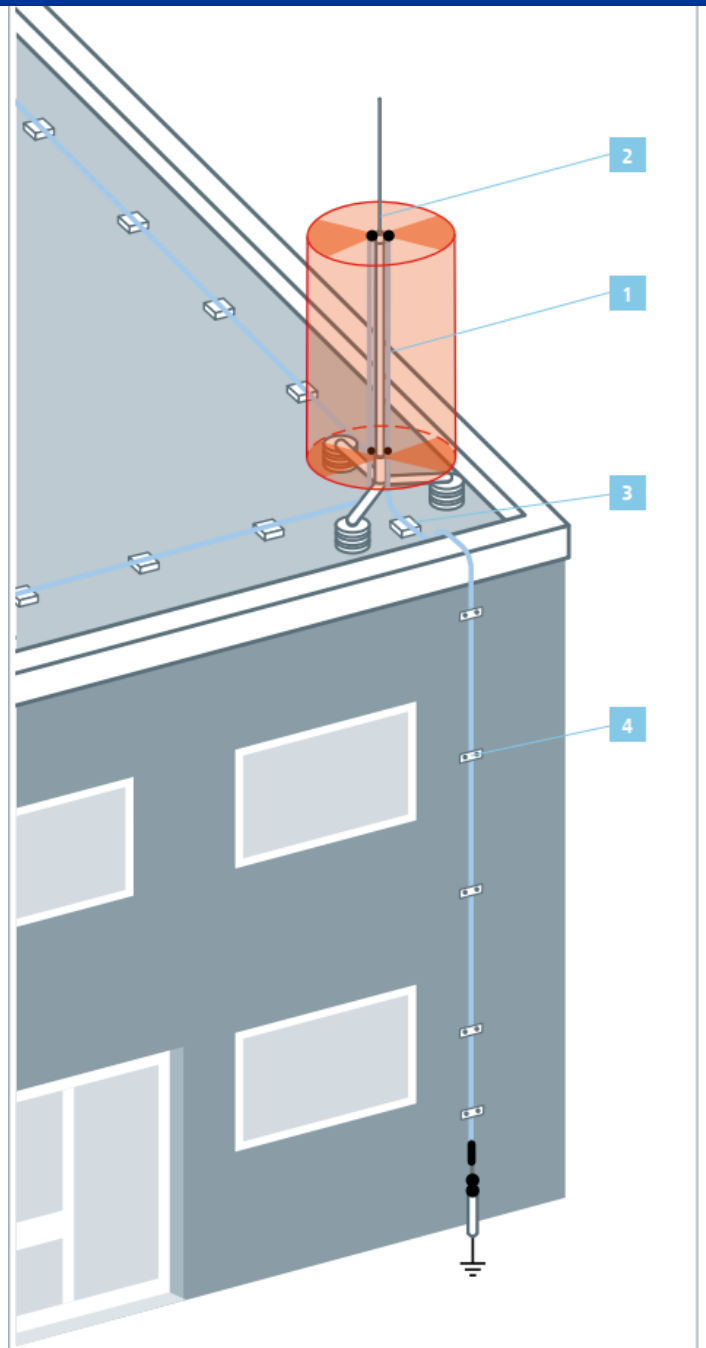
Penangkal petir atap datar dengan PLTS



Lightning Protection System for an Administration Building with solar cell panel



Isolated Lightning Protection System for an Administration Building



Example of a “partly isolated” lightning protection system





Source: Victoria-Blitzschutz GmbH, Lünen

a



a



3.2. FIXED STANDING PV

Panel Surya



PV Plant di tempat terbuka terekspose petir



Panel Surya

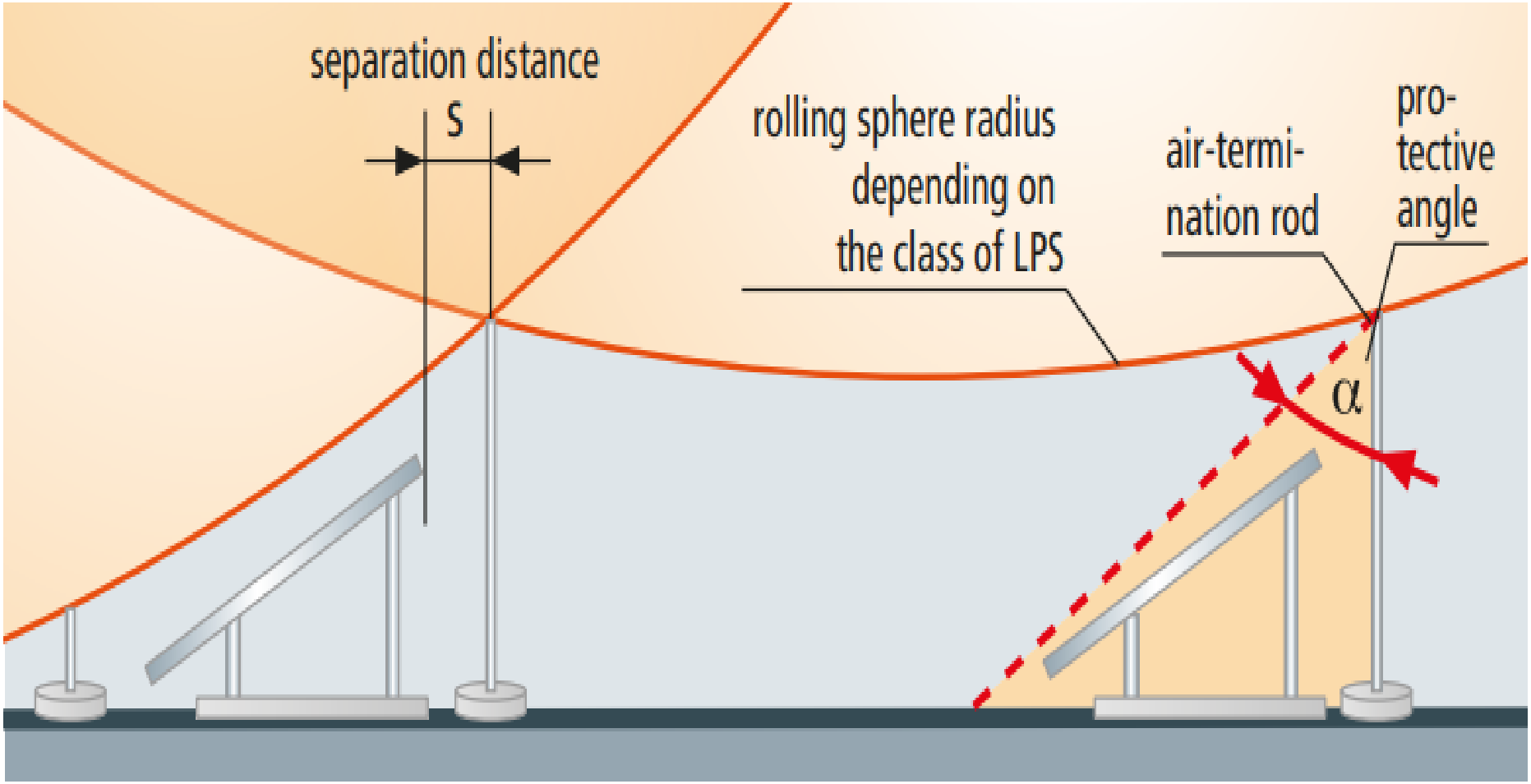


Pemasangan Modul PV menggunakan *Fixed Tilt System*





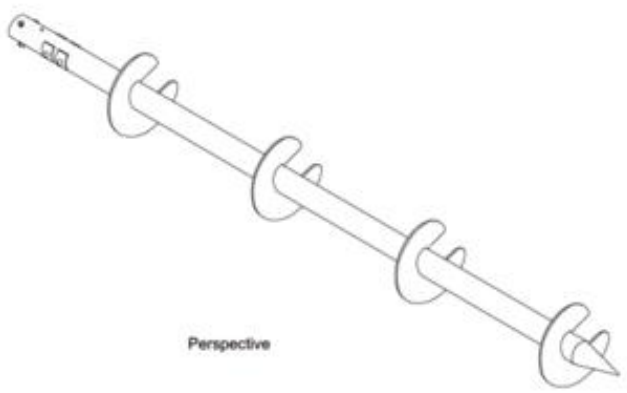
Rolling sphere methode

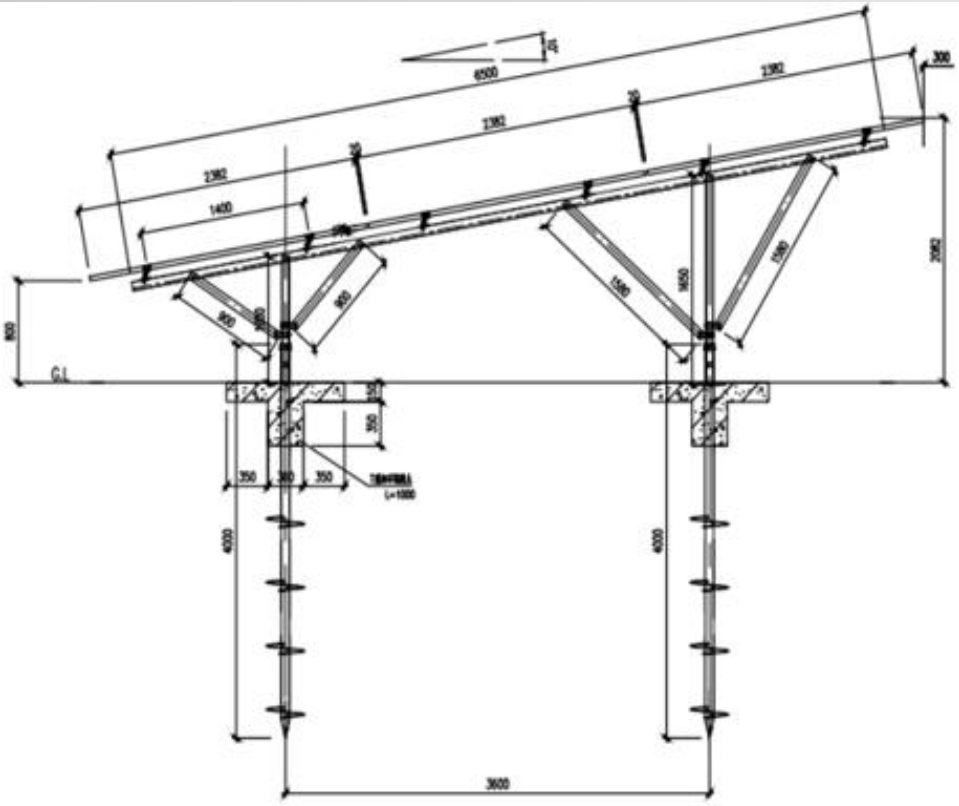


Konstruksi PV Plant

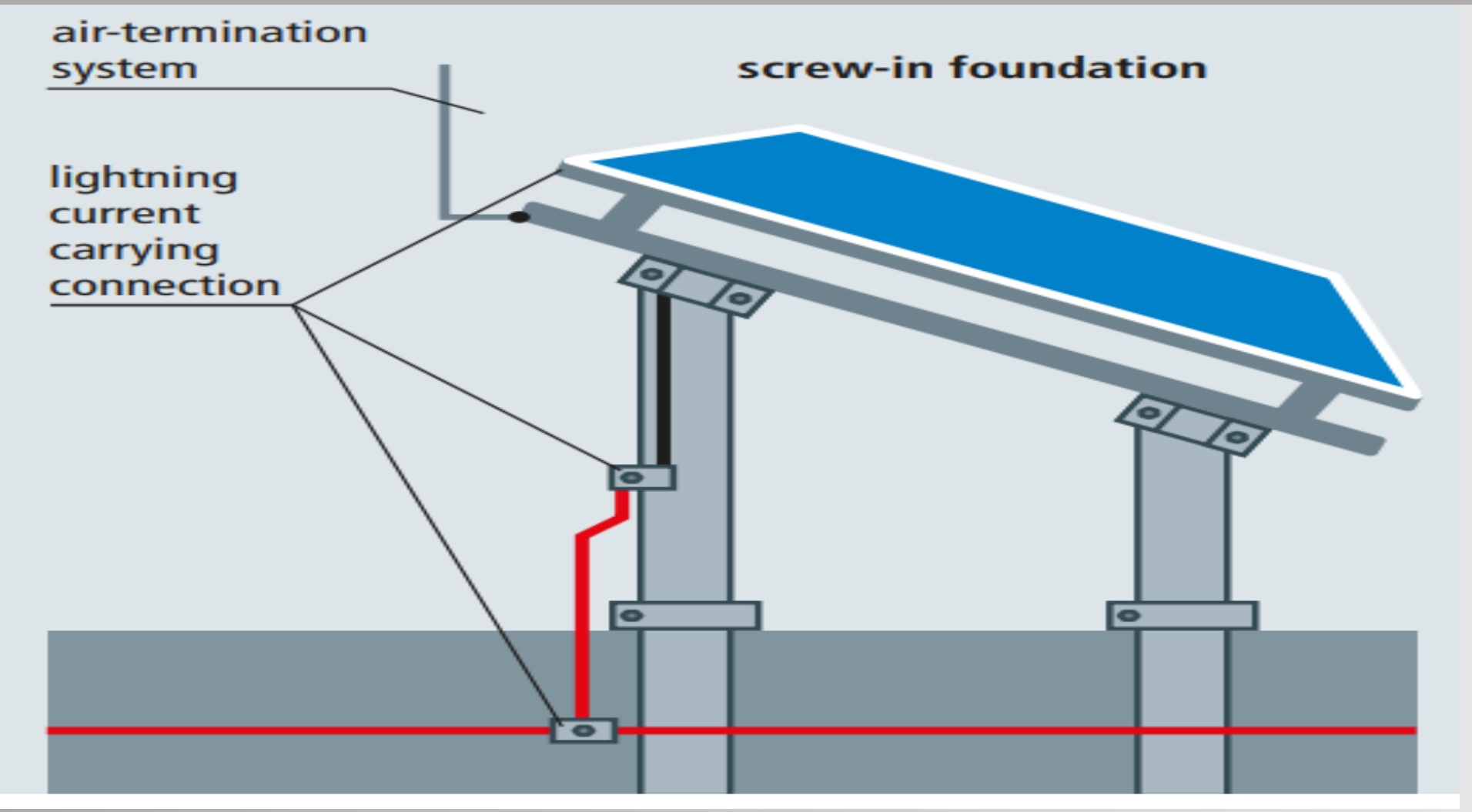


Konstruksi PV Plant

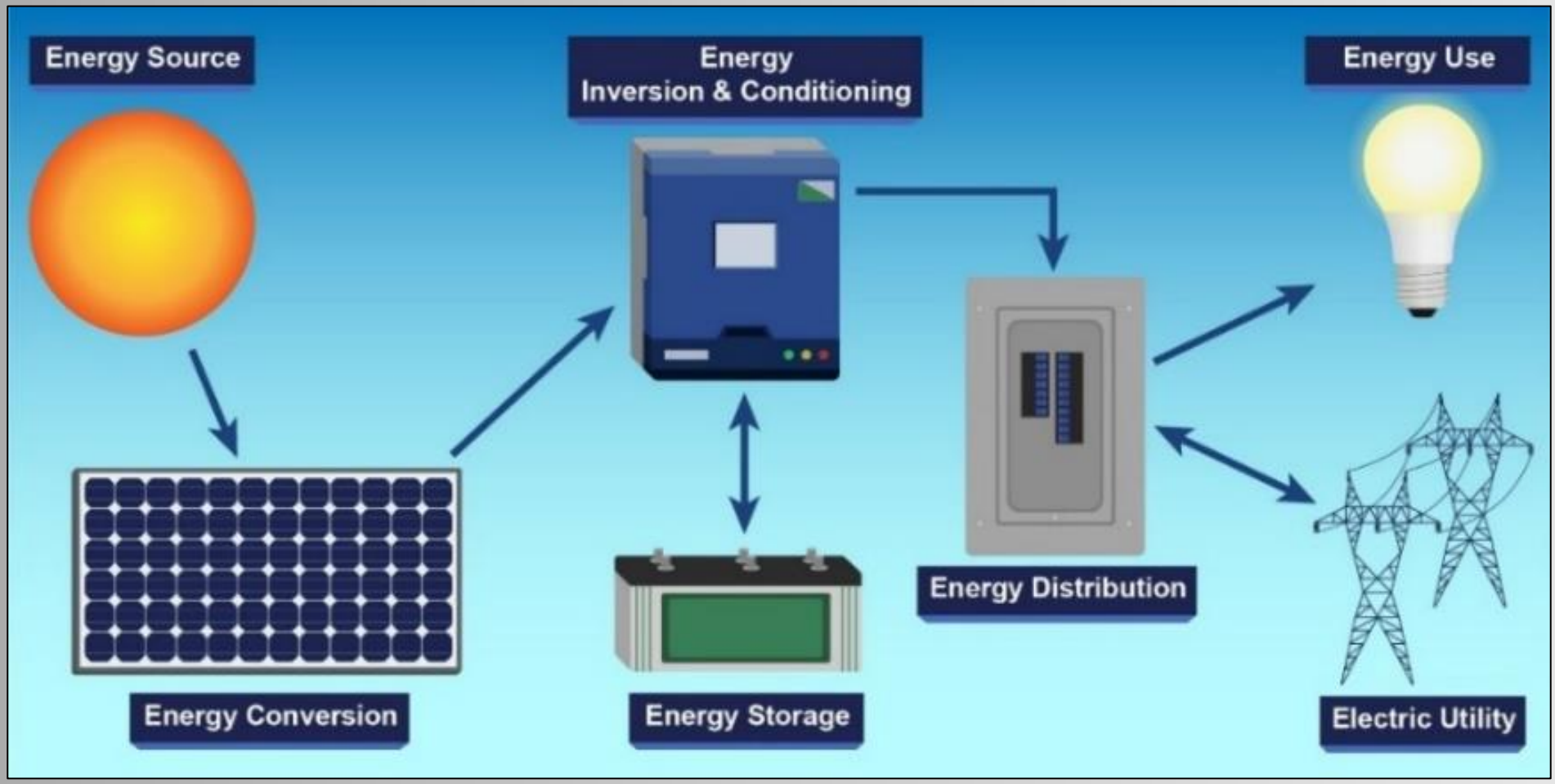




Air Terminal PV Plant



Ilustrasi Sistem PLTS

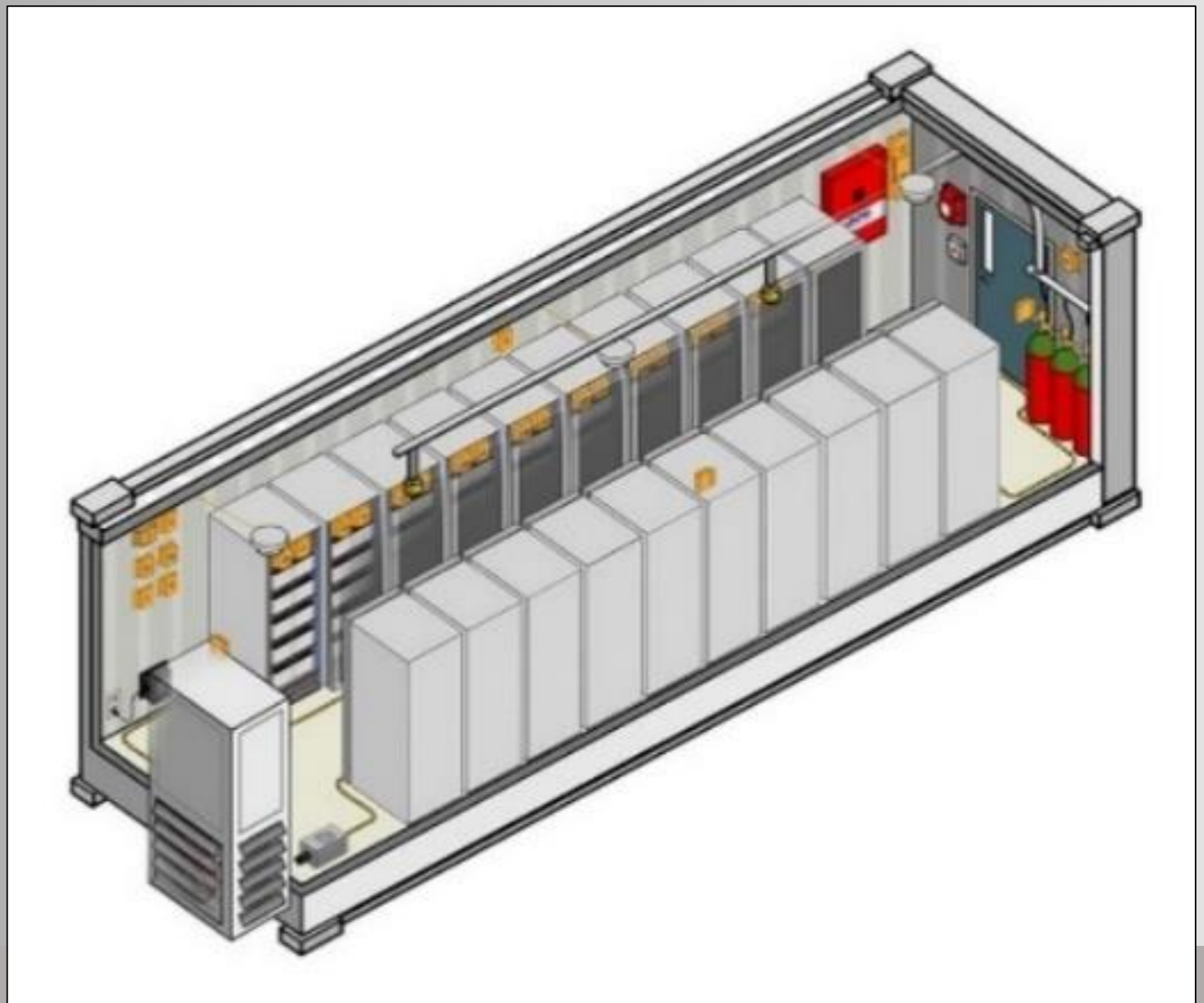


Ilustrasi *Battery Energy Storage System* (BESS)





Ilustrasi BESS Room



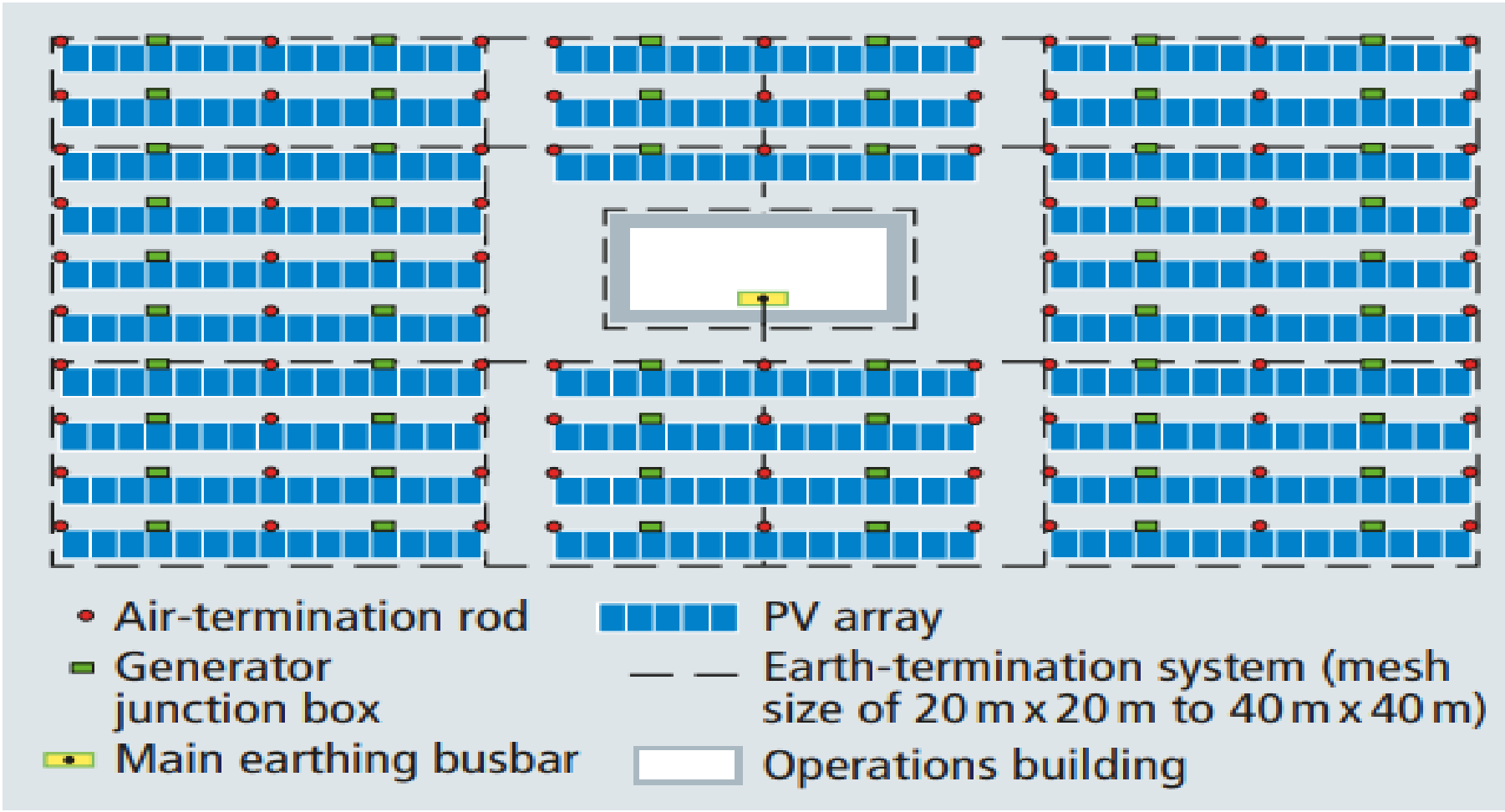
Ilustrasi *Surge Protection Devices*

Surge protection devices (SPD) berfungsi memberikan perlindungan terhadap lonjakan listrik termasuk yang disebabkan secara langsung dan tidak langsung oleh petir.

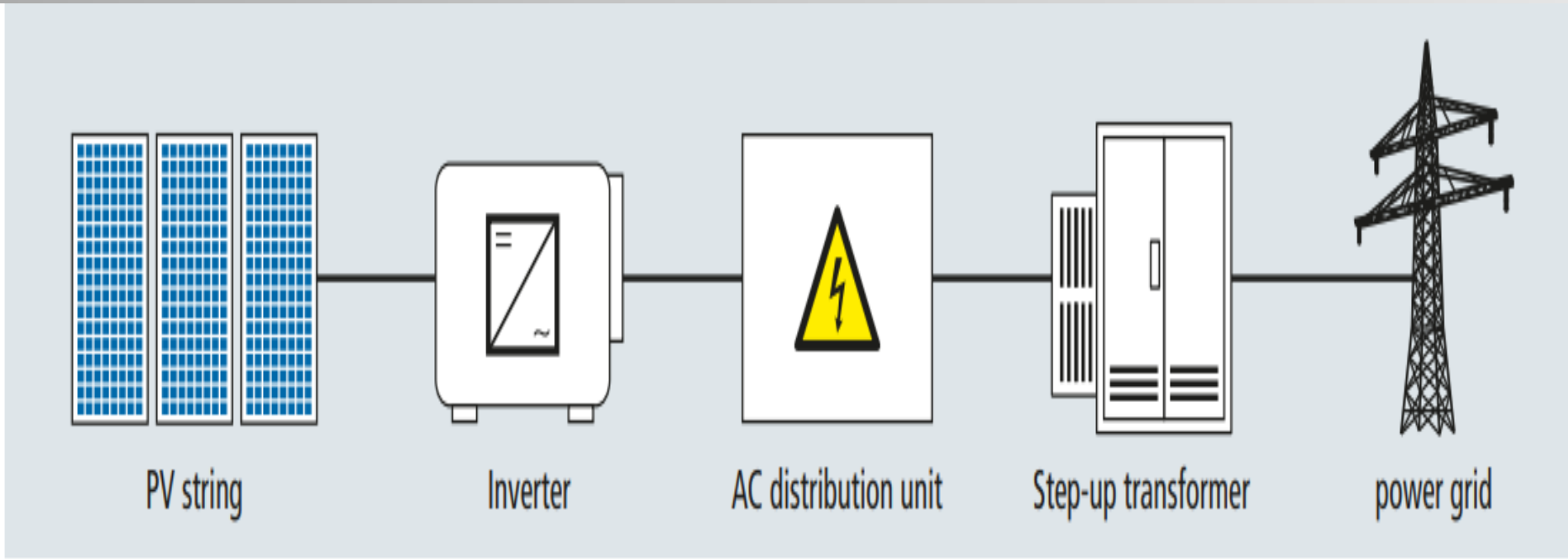


Earth termination system, grounding system

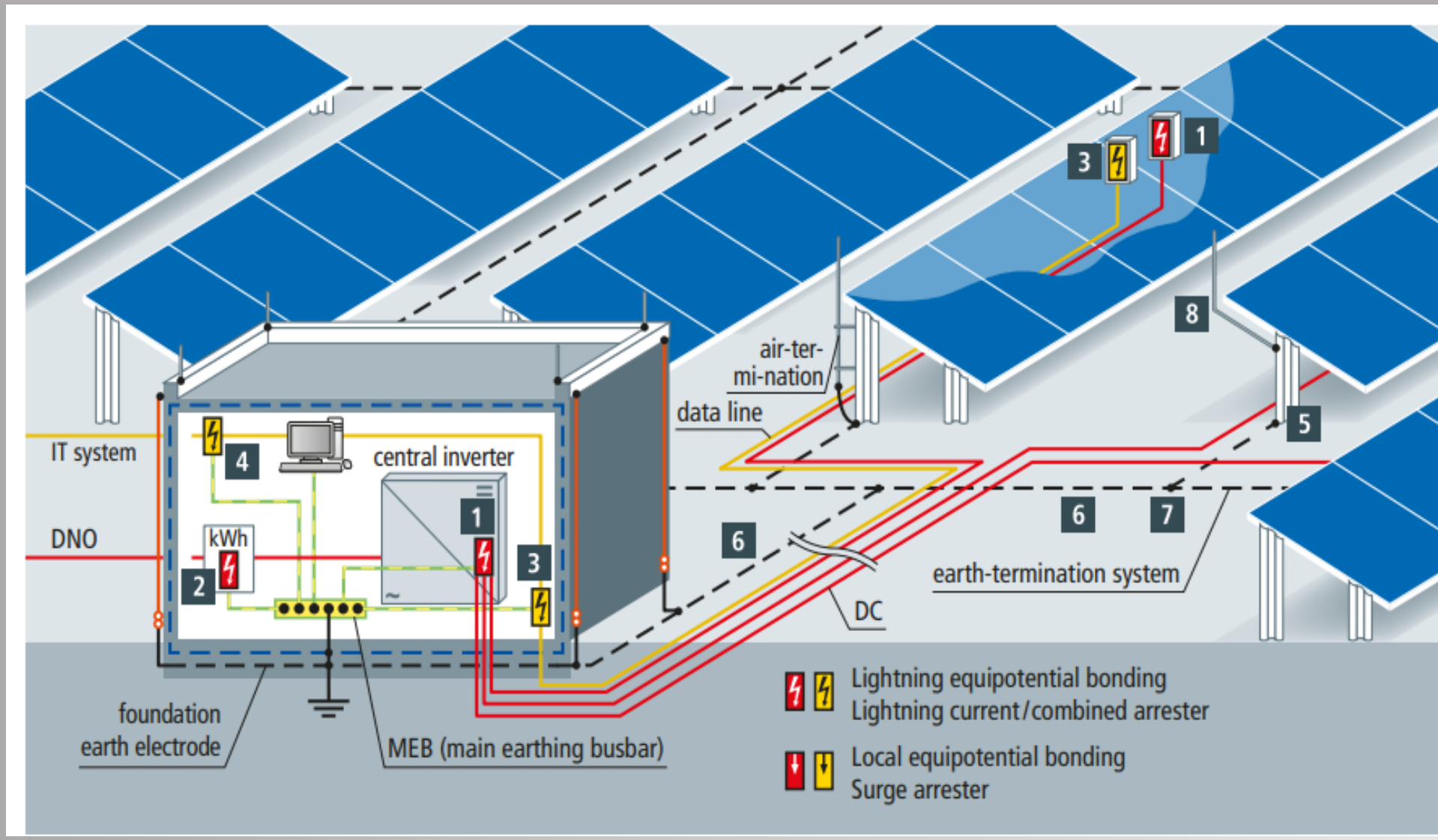
Earth Termination System adalah bentuk dasar implementasi proteksi petir dan surja petir yang efektif, dengan meshed, jejaring, stainless steel, besi galvanis, 10 mm, ukuran 20 m x 20 m sampai 40 m x 40 m, yang digelar ditanam di area PV-Plants dan Control Center. Tahanan Pentanahan RA direkomendasikan kurang dari 10 Ohm.



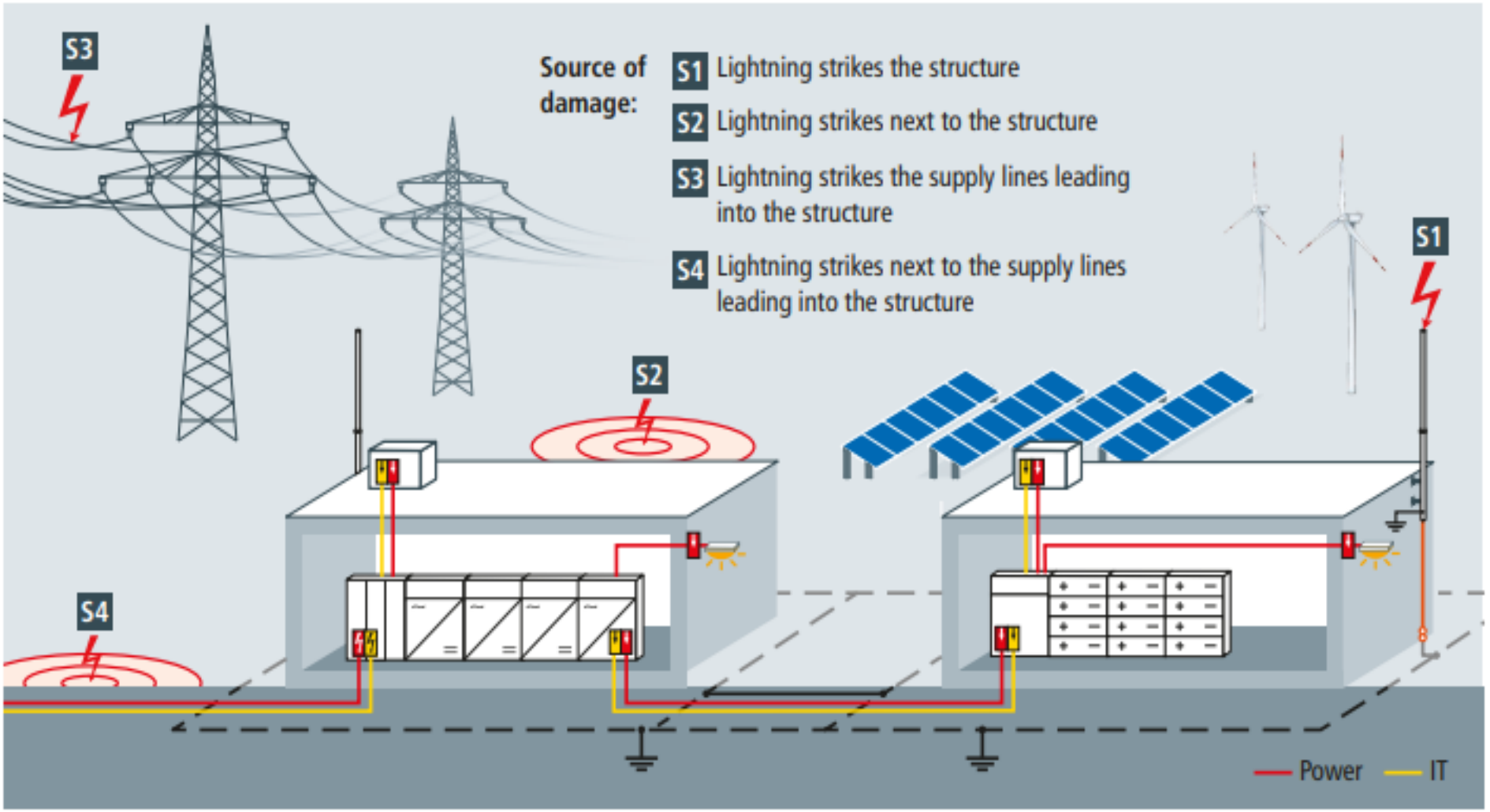
Systematic Structur of Free-Field PV System with String Inverter



PV System and central inverter



Sistem proteksi petir terpasang menentukan keandalan dan keselamatan Plant



Ilustrasi Battery Energy Storage System (BESS)





Contoh Penempatan BESS



Tipikal Tower Transmisi



Gantry Tower

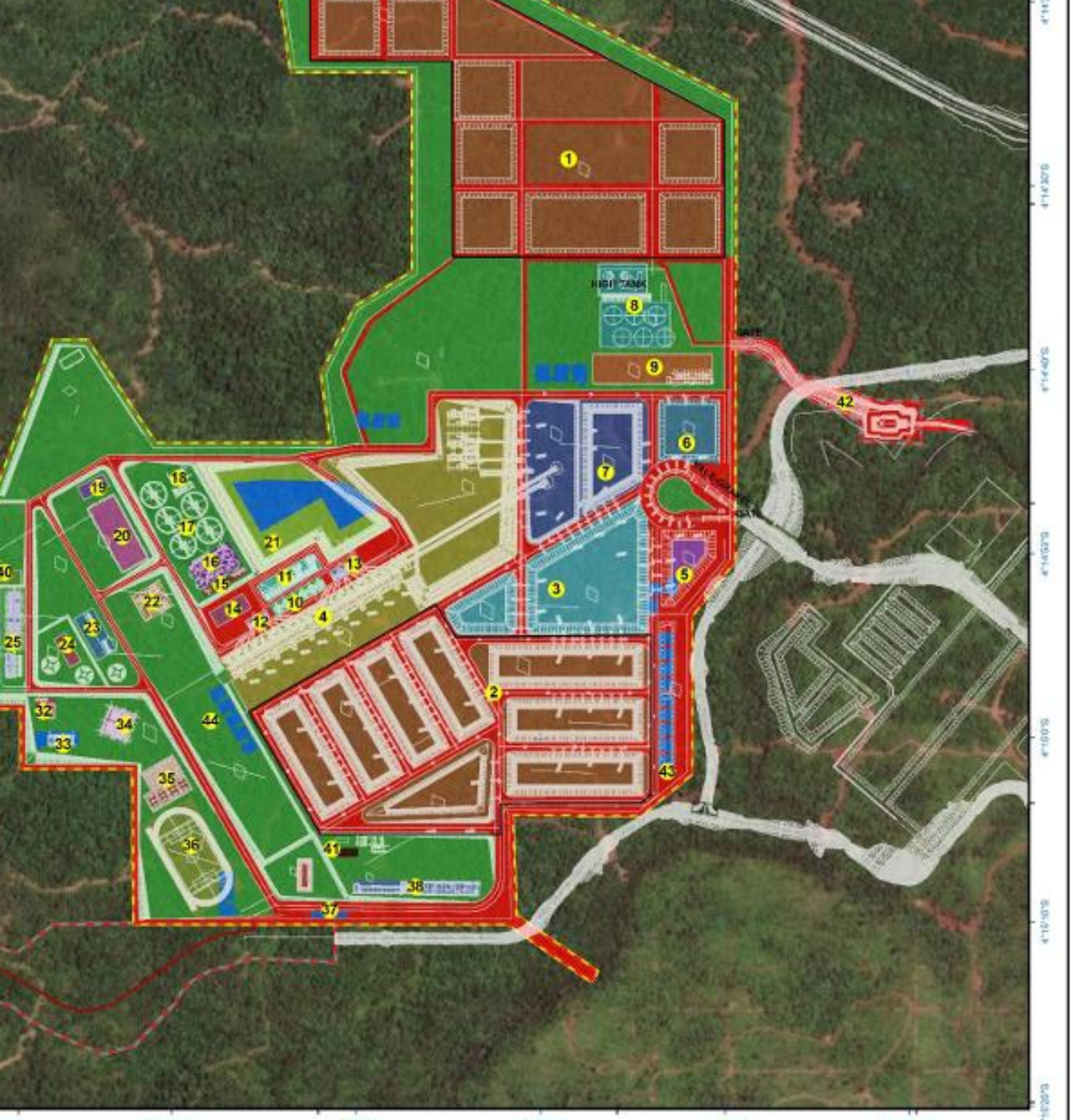


Lattice Tower



Monopole Tower

3.3. FLOATING FIELD PV



U

Skala 1:8.000

0 62,5 125 250 375

Proyeksi : Transverse Mercator
Sistem Koordinat : WGS 1984 UTM Zone 51S
Datum : WGS 1984

© Ukuran Kertas A3

Keterangan

Batas Tapak Proyek FPP

Keterangan

No	Keterangan	No	Keterangan
1	Area Penumpukan Bijih Limonit 1	23	Kantor Pengelolaan Mi
2	Area Penumpukan Bijih Limonit 2	24	Aula Perkantoran
3	Temporary Stock Yard	25	Mess Karyawan Chines
4	Stasiun De-agglomerasi Bijih	26	Mess Karyawan Kantor
5	Area Bongkar Muat	27	Kantin Karyawan Chine
6	Area Penumpukan Kerikil	28	Mess Karyawan Indone
7	Area Pengiriman Kerikil	29	Kantin Karyawan Indon
8	Tangki Penyimpanan	30	Gedung Olah Raga
9	Kolam penampan sampai	31	Supermarket
10	Cyclone feeding tank	32	Genset dan BBC (Fuel
11	Grading and Screening	33	Gereja
12	Tangki penyimpanan Bijih Benih	34	Masjid
13	Tangki penyimpanan Bijih Benih	35	Gedung Olah Raga
14	Gardu Induk Transformer 30kV	36	Lapangan Speek Bola
15	Stasiun Pompa Pencucian	37	Kolam Pengumpul Air
16	Tangki Lumpur Limonit/Limonite Slurry Tank	38	Bengkel Perawatan Ke
17	Stasiun Pengantaran Bubur Bijih Slurry	39	Kolam Pengumpul Air
18	Laboratorium	40	PAL
19	Bengkel Perawatan	41	TPS Limbah B3
20	Gudang	42	Water Intake Station
21	Pompa Daur Ulang Air Produksi	43	Jalan Area Perkerasan
22	Stasiun Pompa	44	Ruang Terbuka Hijau

Sumber :

- Peta Dasar Rupa Bumi Indonesia Badan Informasi Geospasial Skala 1:25.000 (2018)
- Peta Tematik Zhejiang Industri Design and Research Institute Co., Ltd.
- Basemap Google Earth Online (2023)



greencorp

Anchor construction





Transformer and inverter lifting installation water



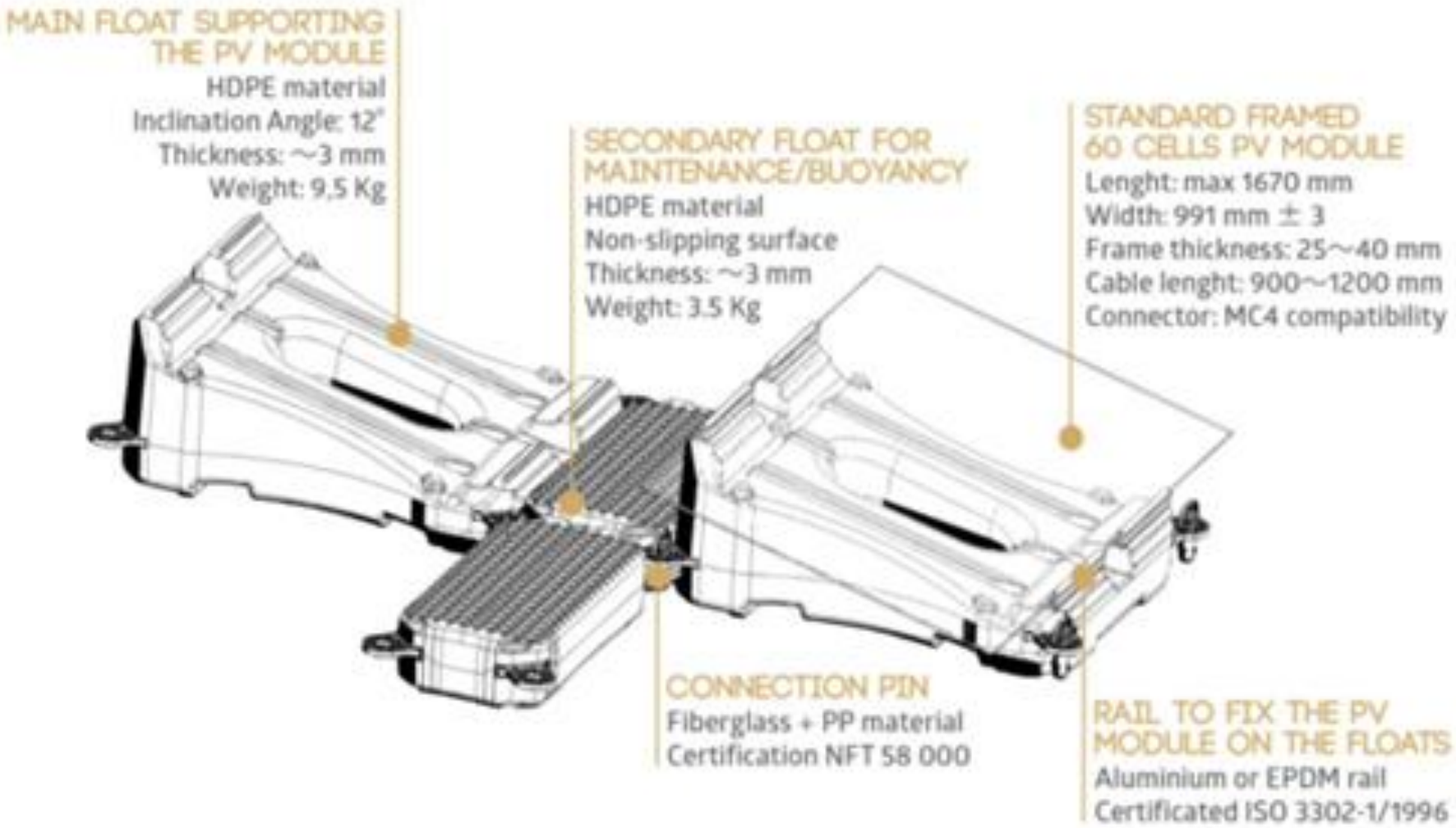




Lifting to the water pontoon

Struktur Floater HDPE untuk Pembangkit Listrik Terapung

Mooring Arrangement



Typical shore temporary assembly platform layout

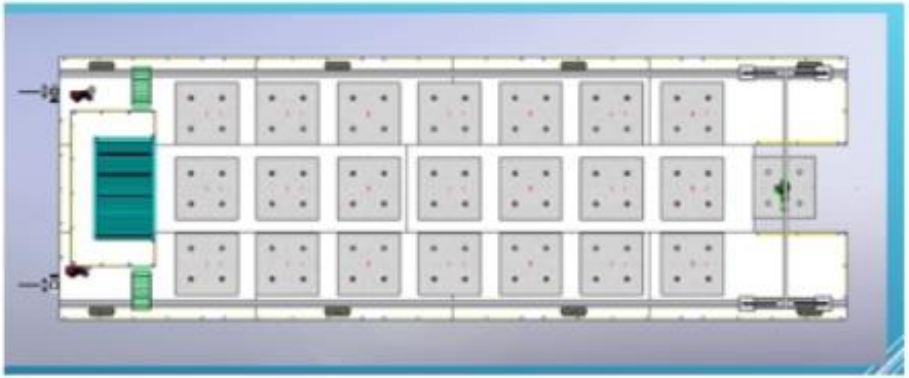


Ilustrasi Perakitan Modul Surya Terapung di *Assembly Line*

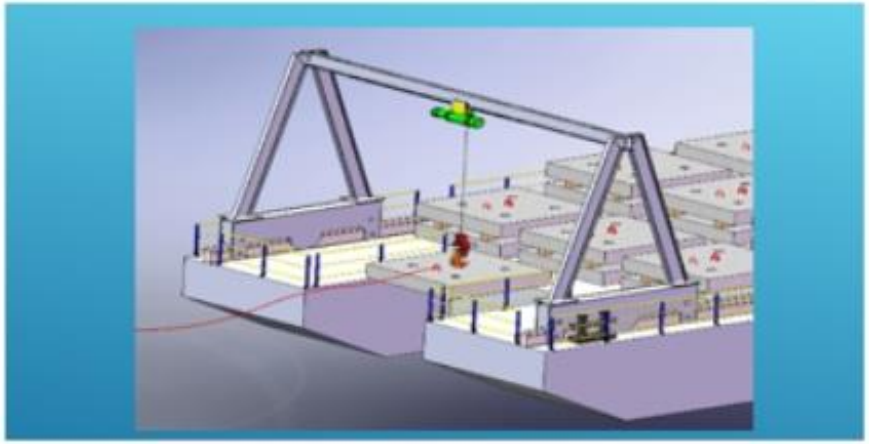




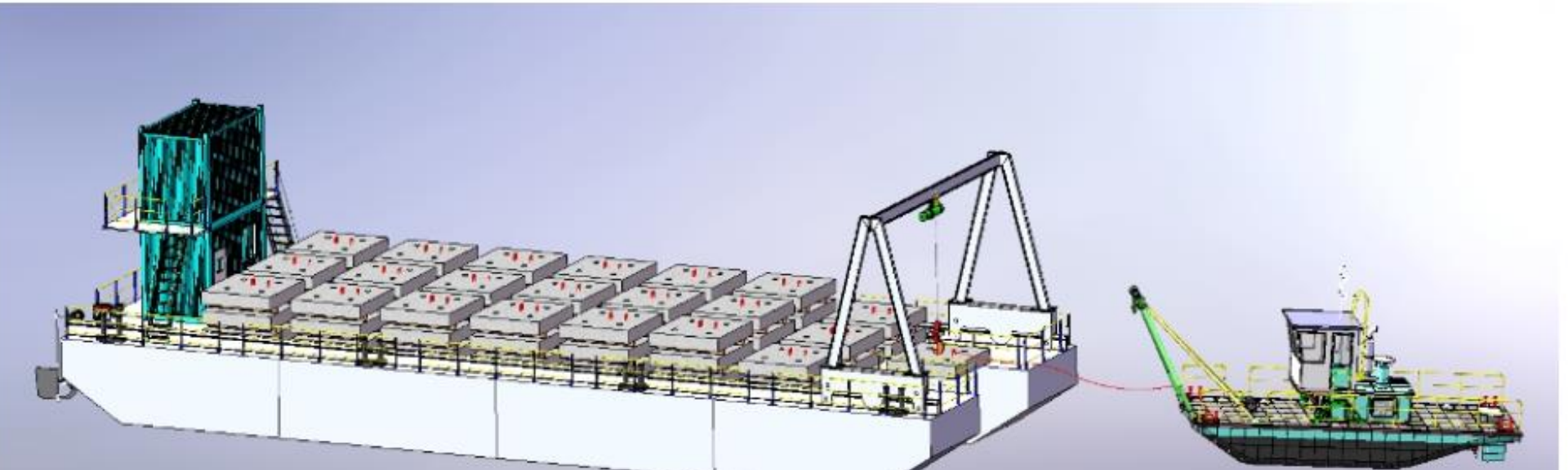
Anchor and boat



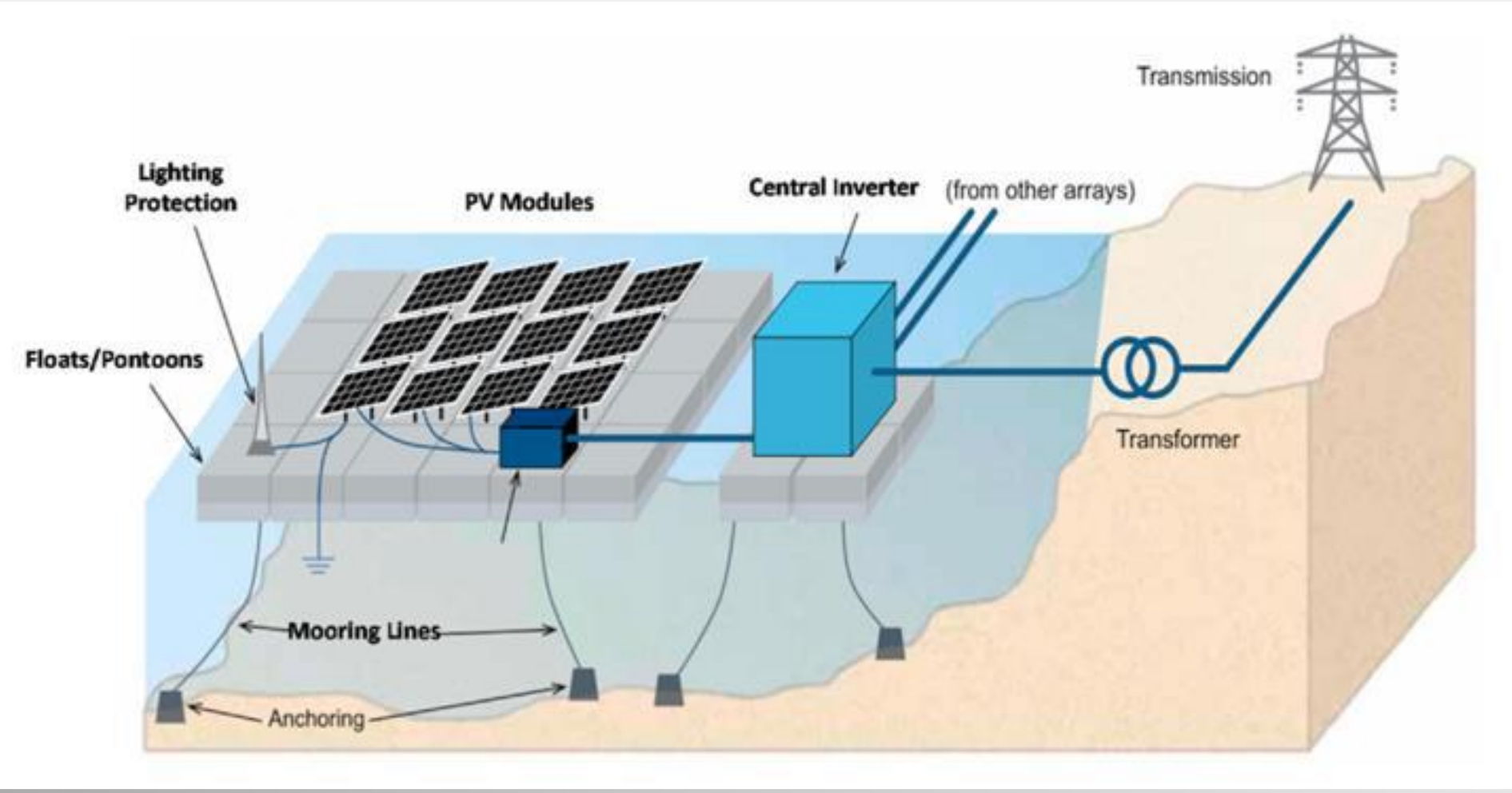
Total area: about 372 m
Loadable area: ~ 260 m
Design loading capacity: 3 rows vertically, 9 rows horizontally, each 3 floors (one row of the ship is 2, a total of 60 pieces of paving blocks.



anchor boat configuraLifting weight: 12.5 tons
(maximum) track length: 30.0 meters
Span: 10.0m
Lifting height: 110.0m (with automatic decoupling and anti-rotation sleeve)tion



External Lightning Protection PLTS Terapung



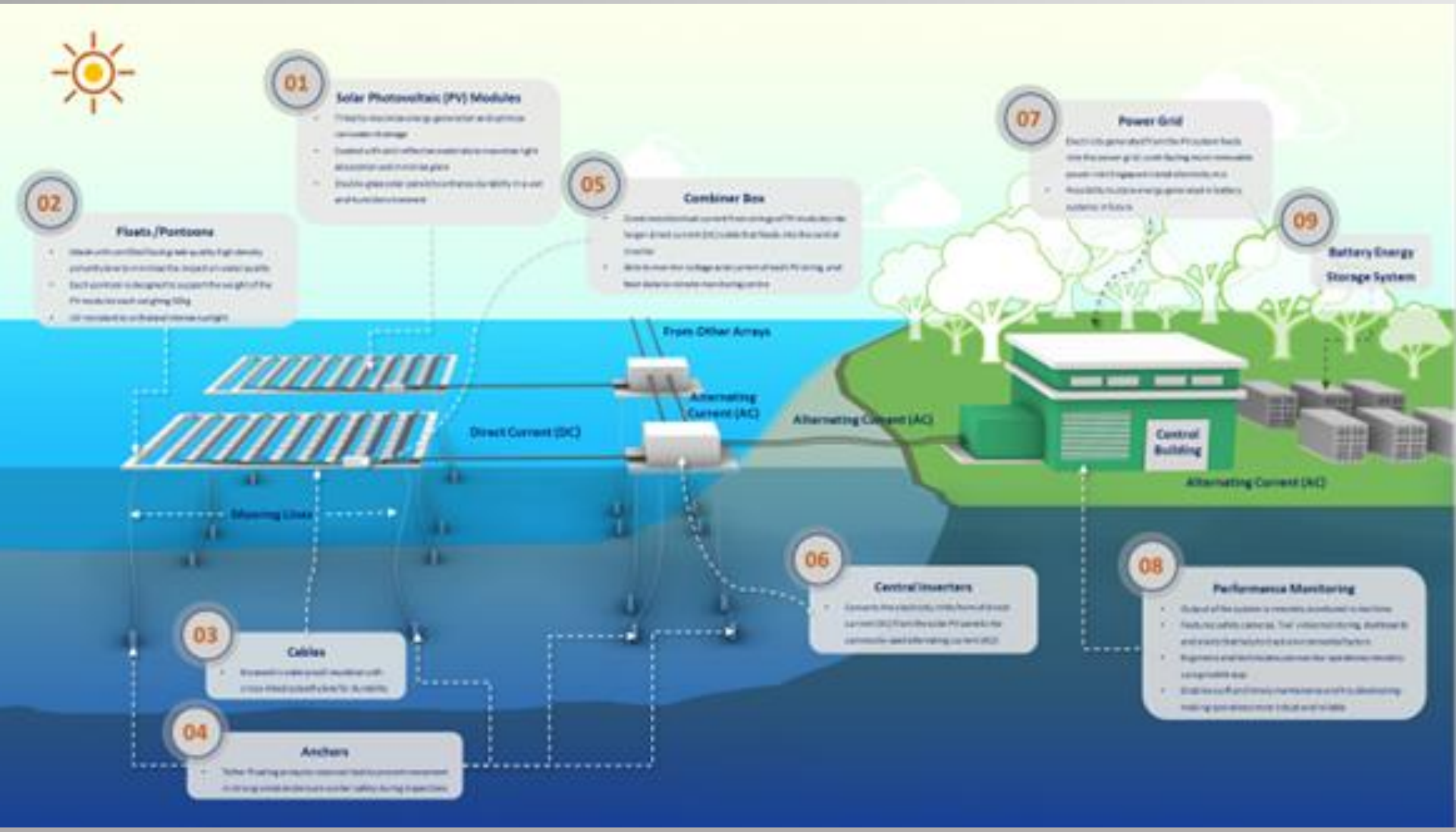
External Lightning Protection meliputi:

- Pemasangan dan Penempatan Free standing tower air terminal
- Perbaikan sistem grounding frame photo voltaic panel
- Bonding

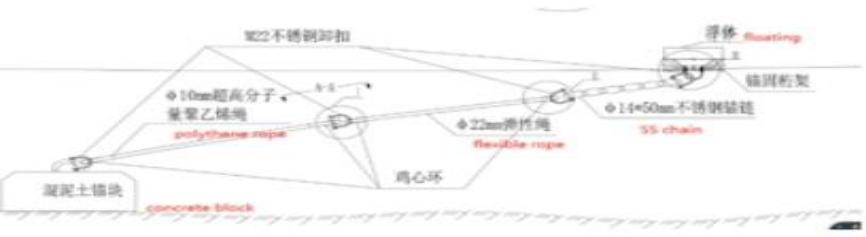
Mineral Oil, Silicon Oil

Minyak trafo sebagai isolasi dan pendingin Trafo Daya

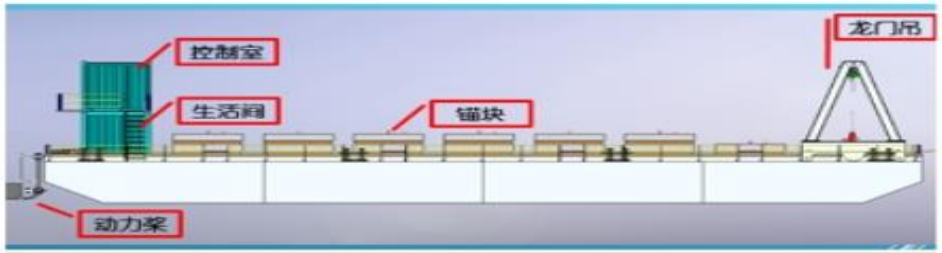
Penggunaan minyak trafo tipe askarel, jenis mineral oil, cairan silicon sangat toxid yang berbahaya mencemari air danau yang digunakan sebagai bahan baku air minum



Anchor system construction



Anchor in water



Total length of platform: 34.0m
Total platform width: 11.4m
Platform height: 2.5m (excluding control room height)
Full load displacement: 250t

anchor boat configuration

Terima kasih